AWS CloudWatch Monitoring with SNS Email Alerts

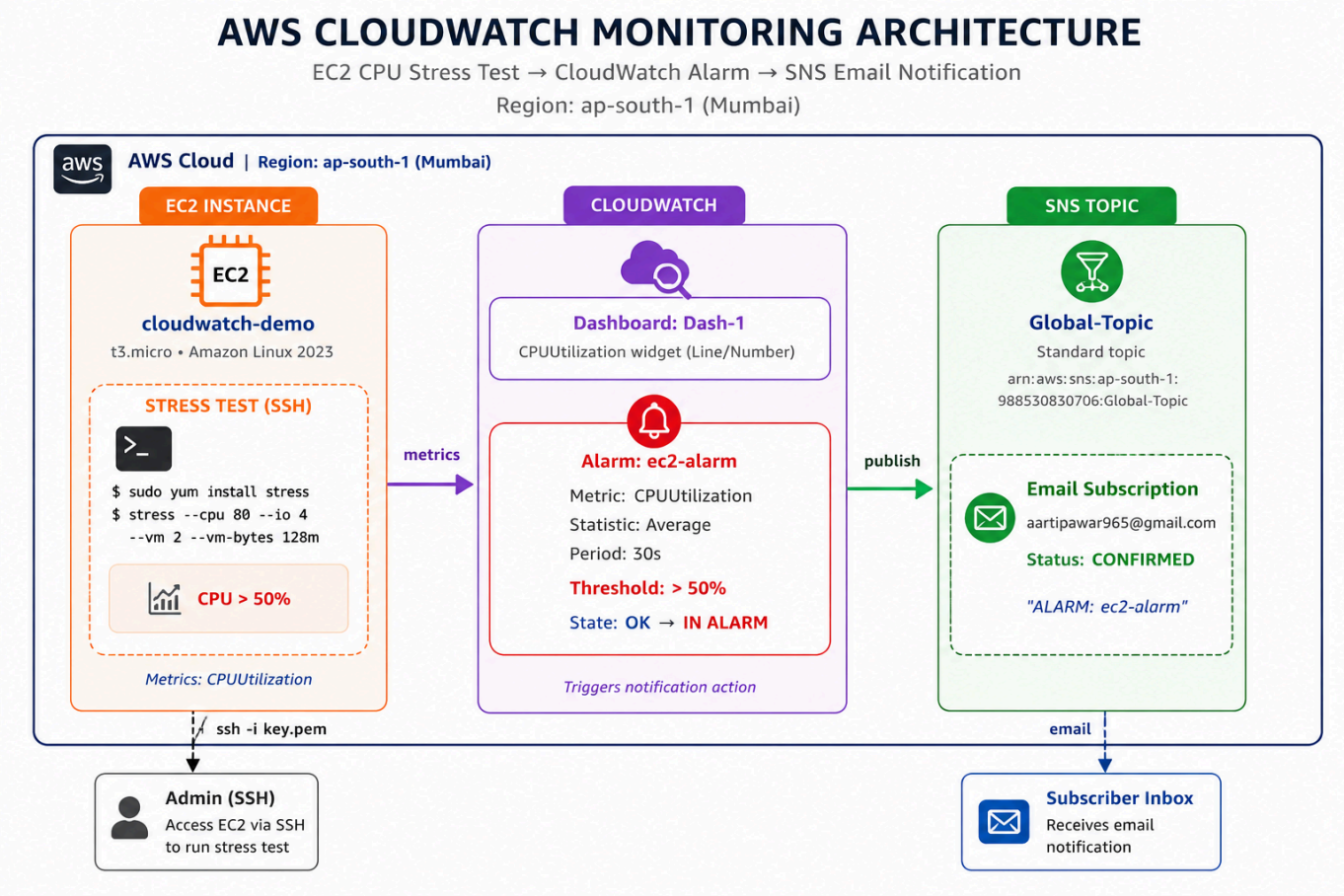
Project Overview

This project demonstrates how to build a complete monitoring and alerting solution using Amazon CloudWatch, CloudWatch Agent, and Amazon SNS. The solution continuously monitors the health and performance of an Amazon EC2 instance by collecting both default AWS metrics and operating system-level metrics such as CPU, Memory, Disk, and Network utilization.

A custom CloudWatch Dashboard was created to visualize system performance in real time. CloudWatch Alarms were configured to detect high CPU utilization, and Amazon SNS was integrated to automatically send email notifications whenever the alarm threshold was exceeded. A CPU stress test was performed on the EC2 instance to validate the monitoring pipeline and ensure that alerts were triggered successfully.

Architecture Overview

The monitoring workflow begins with an Amazon EC2 instance running the CloudWatch Agent. The agent collects operating system metrics and sends them to Amazon CloudWatch. A custom dashboard displays real-time metrics, while CloudWatch Alarms continuously monitor CPU utilization. When the configured threshold is exceeded during a stress test, the alarm changes to the ALARM state and automatically publishes a notification to an Amazon SNS topic. SNS then sends an email notification to the subscribed recipient, enabling immediate awareness of performance issues.



Project Objectives

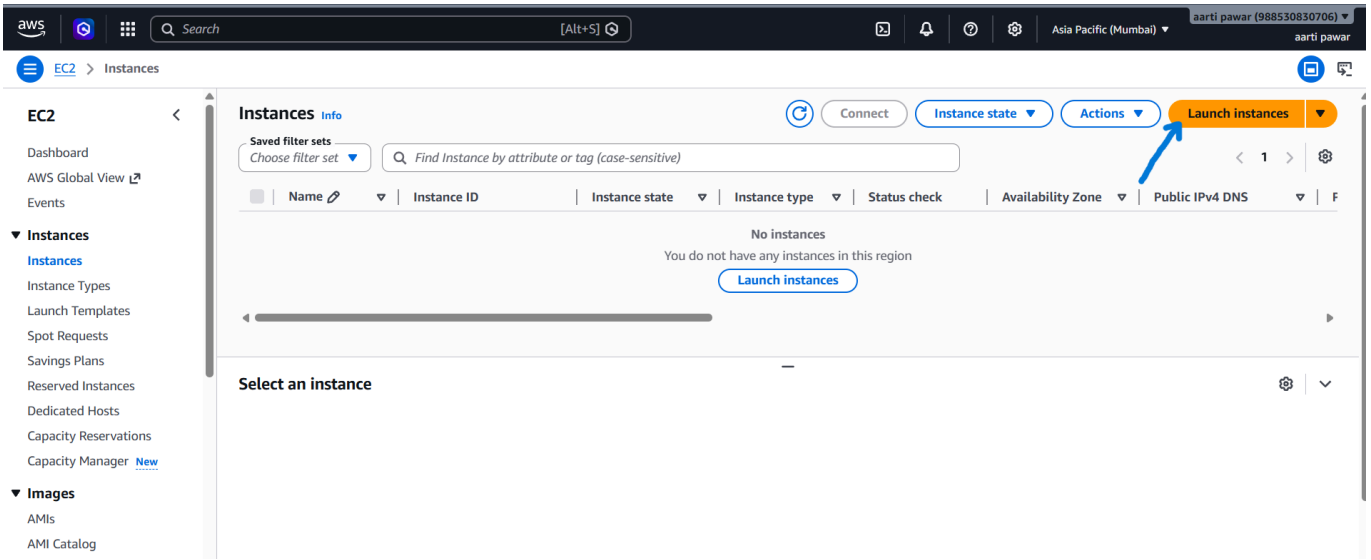
- Monitor Amazon EC2 performance using Amazon CloudWatch.
- Install and configure the CloudWatch Agent.
- Collect CPU, Memory, Disk, Network, and Disk I/O metrics.
- Create a custom CloudWatch Dashboard.
- Configure CloudWatch Alarms for CPU utilization.
- Integrate Amazon SNS for email notifications.
- Perform CPU stress testing to validate alarms.
- Enable proactive infrastructure monitoring.

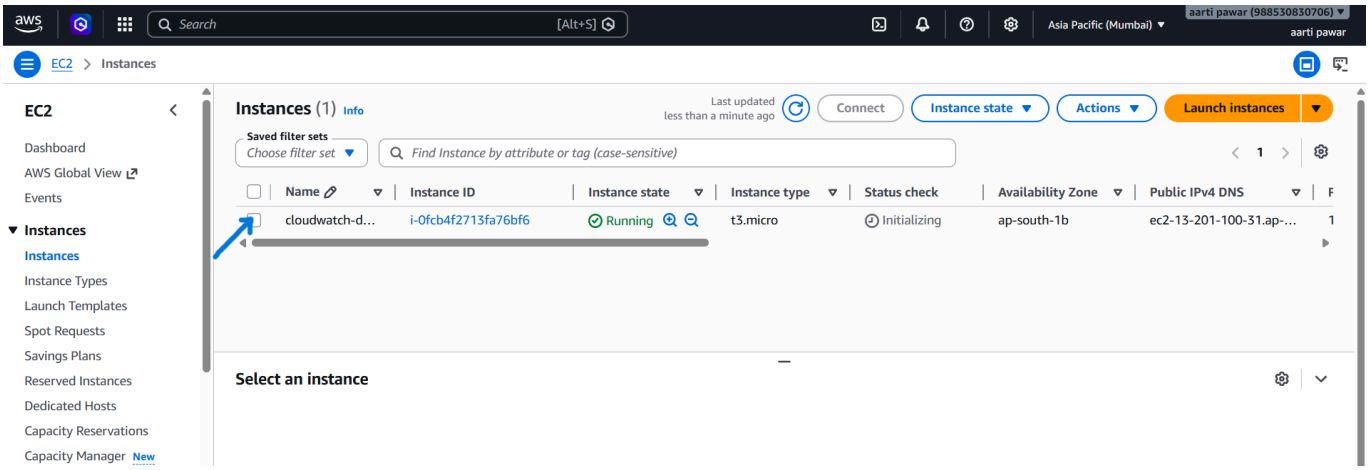
🛠 AWS Services Used

- Amazon EC2
- Amazon CloudWatch
- Amazon CloudWatch Agent
- CloudWatch Dashboard
- CloudWatch Alarms
- Amazon SNS
- AWS Systems Manager Parameter Store
- IAM
- Amazon Linux 2023

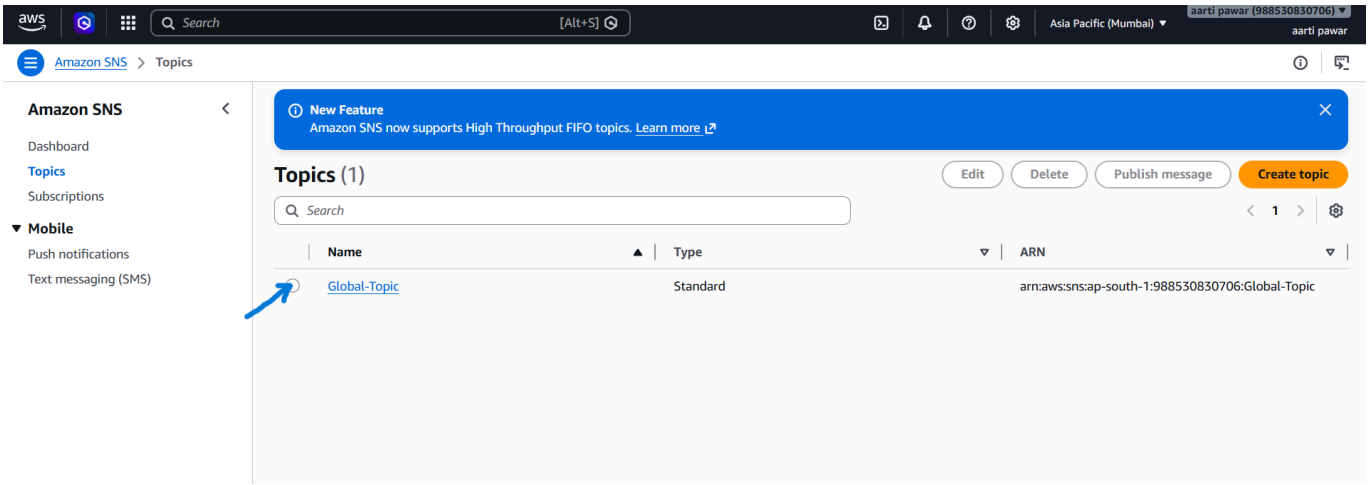
⚙️ Implementation Steps

Launch an Amazon EC2 instance running Amazon Linux.

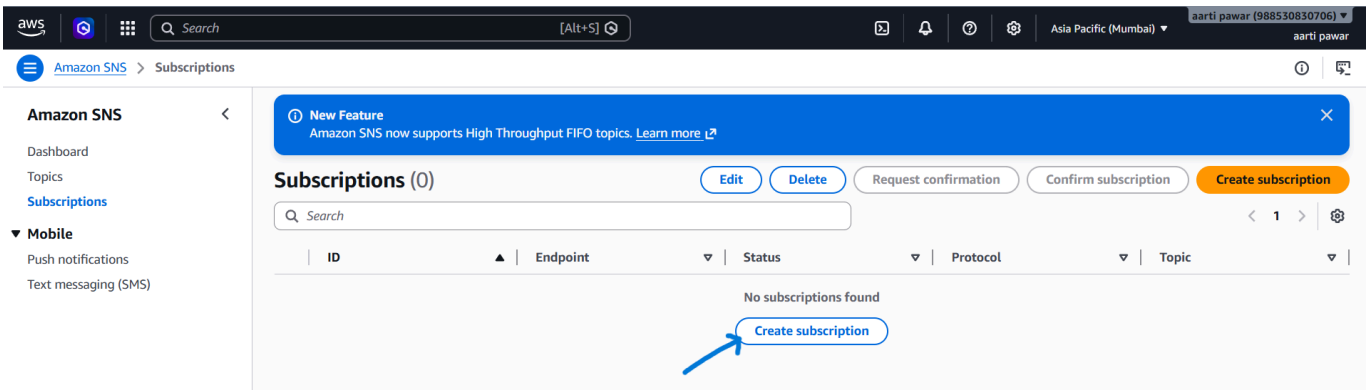


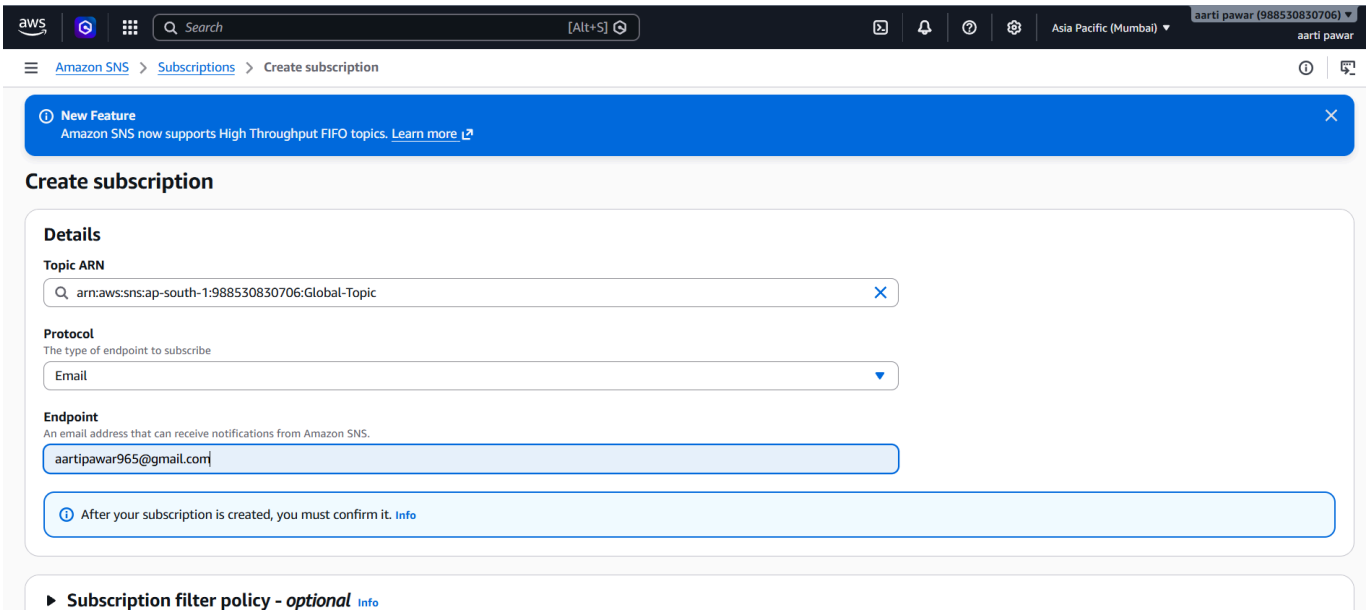


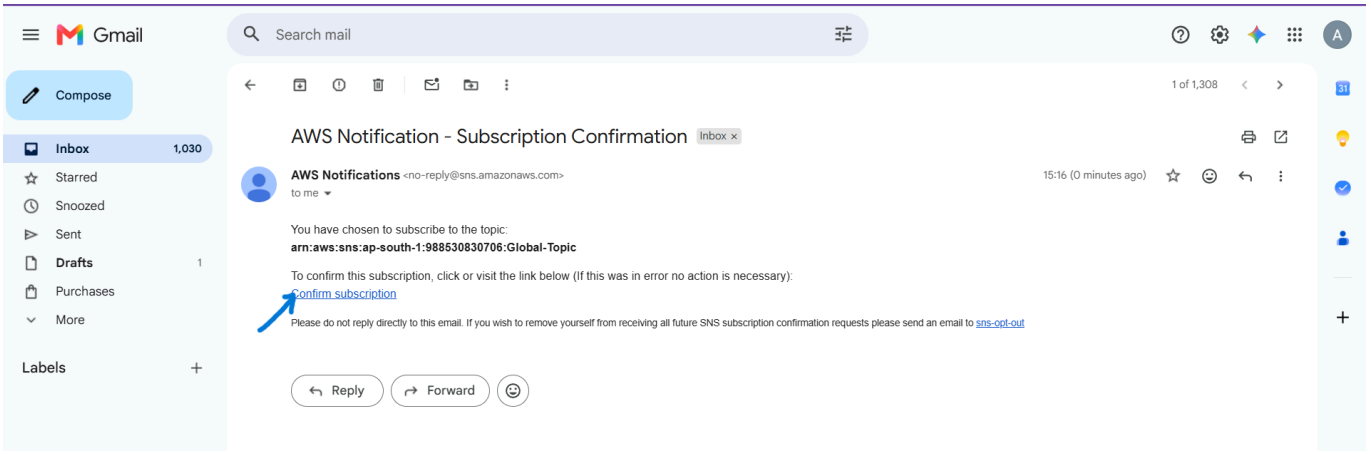
Navigate to Amazon SNS and create a new Standard topic named Global-Topic.

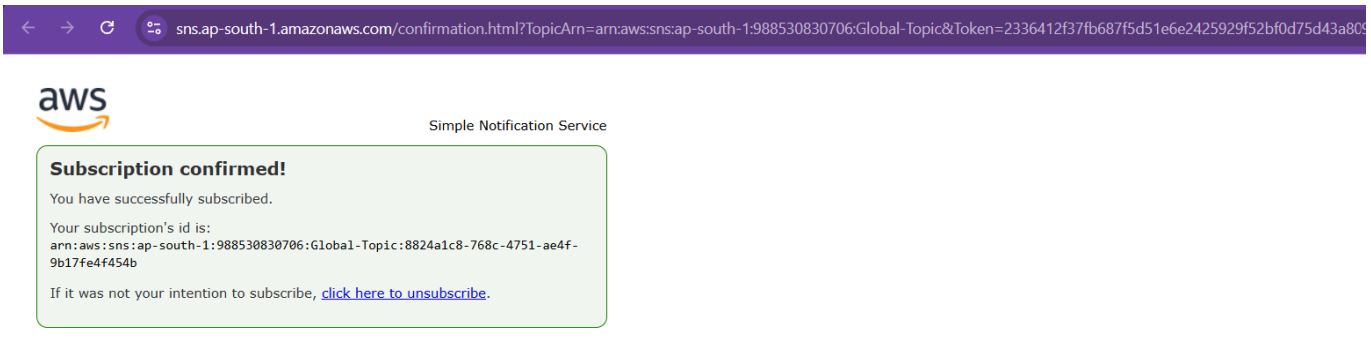


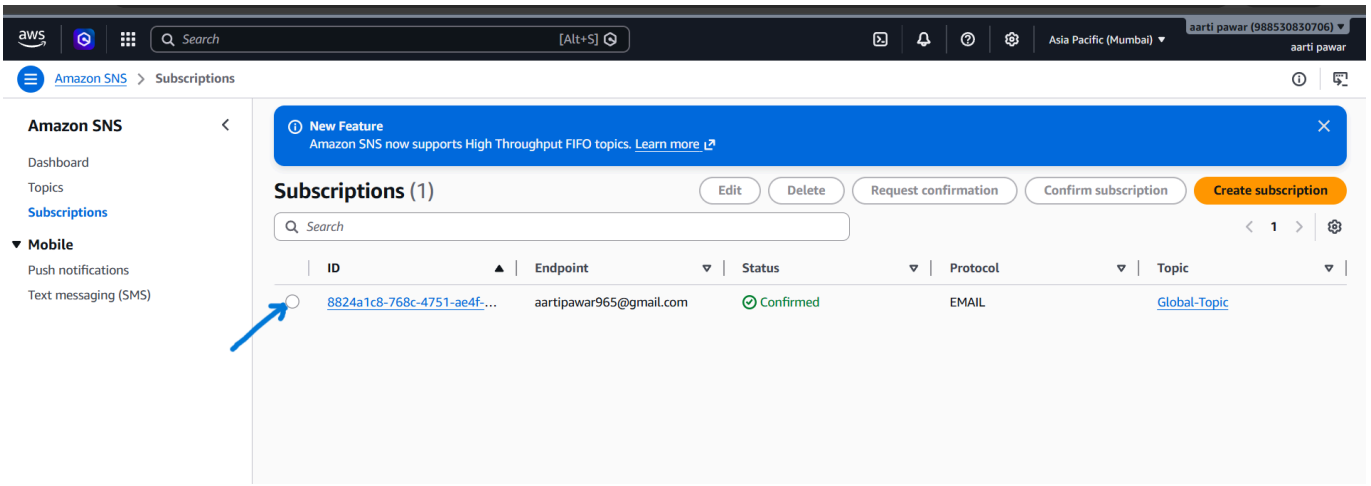
Create Subscription



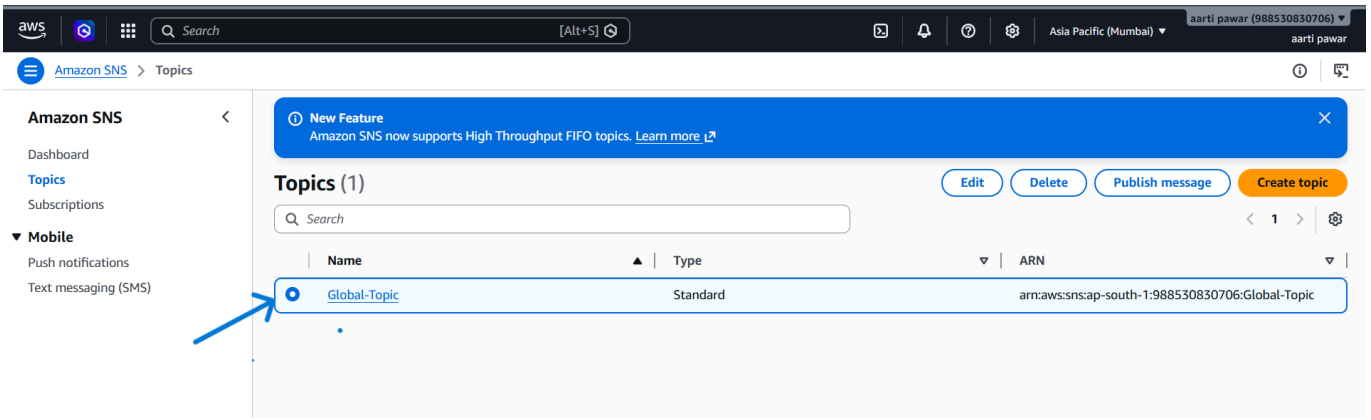

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Amazon SNS > Subscriptions > Create subscription
New Feature
Amazon SNS now supports High Throughput FIFO topics. Learn more
Create subscription
Details
Topic ARN
arn:aws:sns:ap-south-1:988530830706:Global-Topic
Protocol
The type of endpoint to subscribe
Email
Endpoint
An email address that can receive notifications from Amazon SNS.
aartipawar965@gmail.com
After your subscription is created, you must confirm it. Info
Subscription filter policy - optional Info

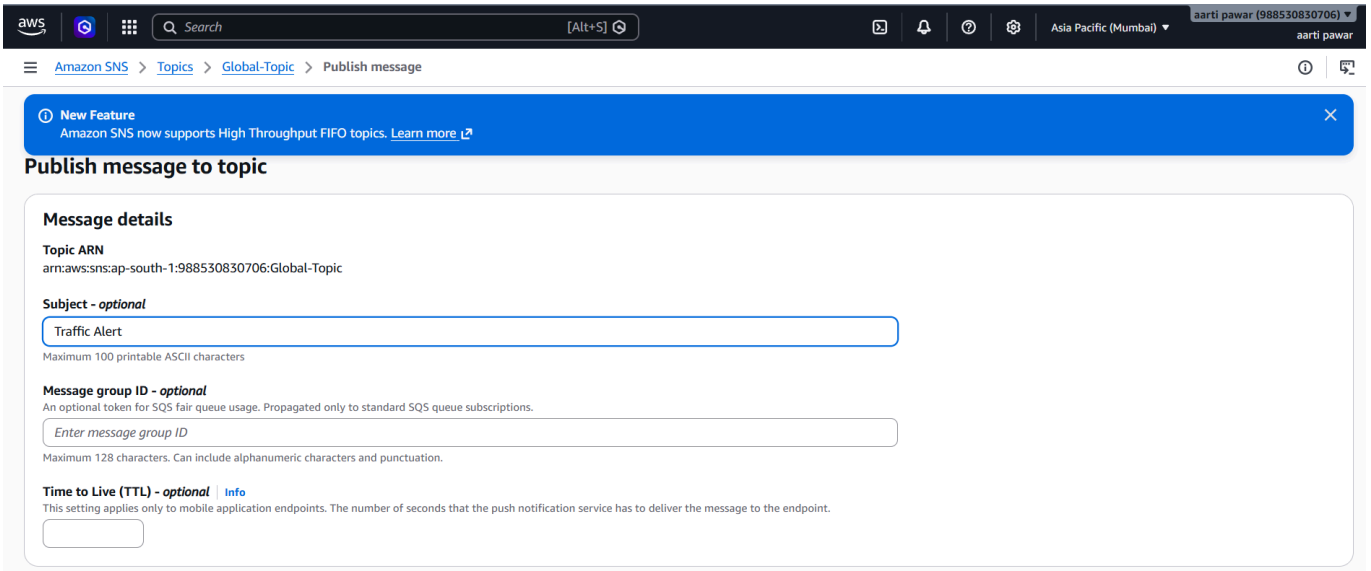

Gmail
Search mail
Compose
Inbox 1,030
Starred
Snoozed
Sent
Drafts 1
Purchases
More
Labels
1 of 1,308
AWS Notification - Subscription Confirmation
Inbox
AWS Notifications <no-reply@sns.amazonaws.com>
to me
15:16 (0 minutes ago)
You have chosen to subscribe to the topic:
arn:aws:sns:ap-south-1:988530830706:Global-Topic
To confirm this subscription, click or visit the link below (If this was in error no action is necessary):
Confirm subscription
Please do not reply directly to this email. If you wish to remove yourself from receiving all future SNS subscription confirmation requests please send an email to sns-opt-out
Reply
Forward

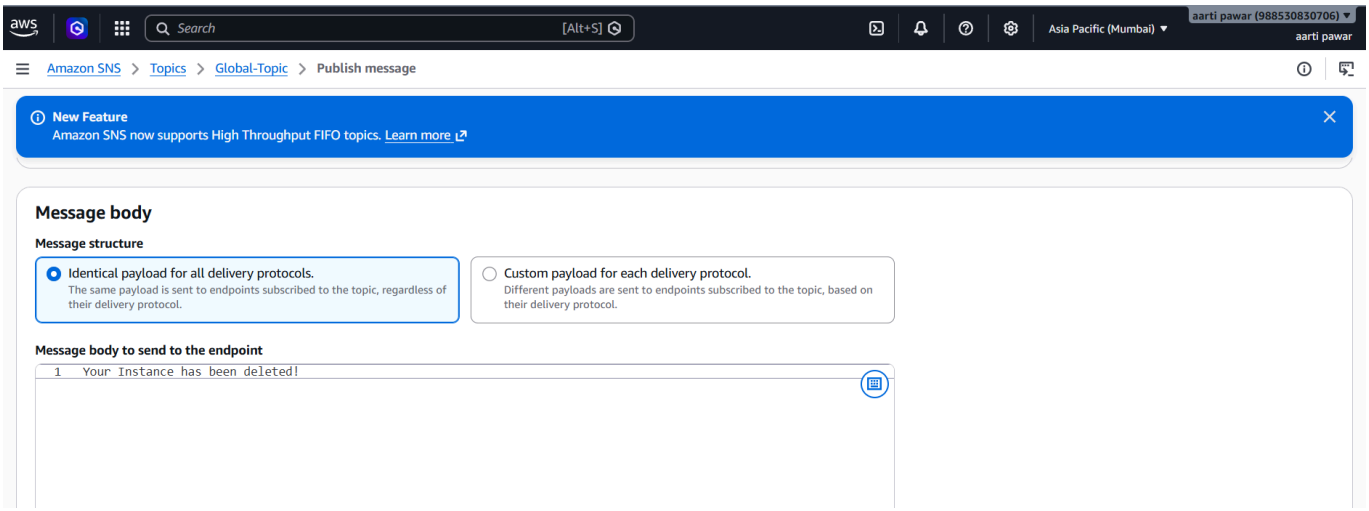

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Simple Notification Service
Subscription confirmed!
You have successfully subscribed.
Your subscription's id is:
arn:aws:sns:ap-south-1:988530830706:Global-Topic:8824a1c8-768c-4751-ae4f-9b17fe4f454b
If it was not your intention to subscribe, click here to unsubscribe.



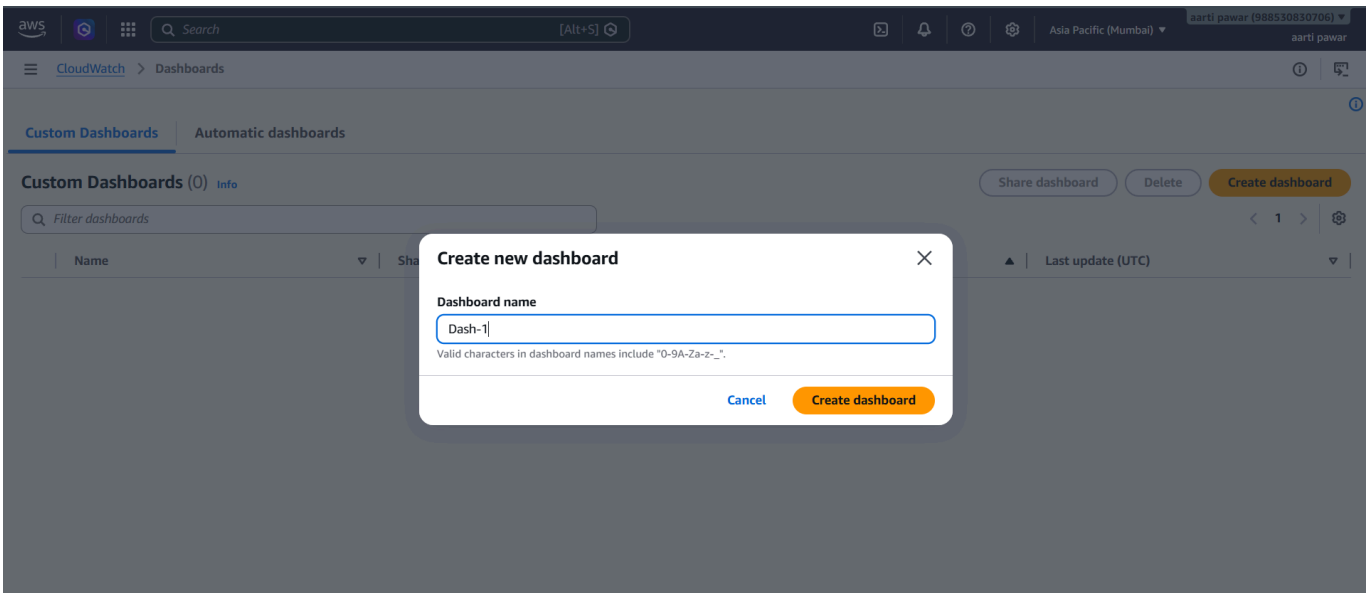
Configure Global Topic

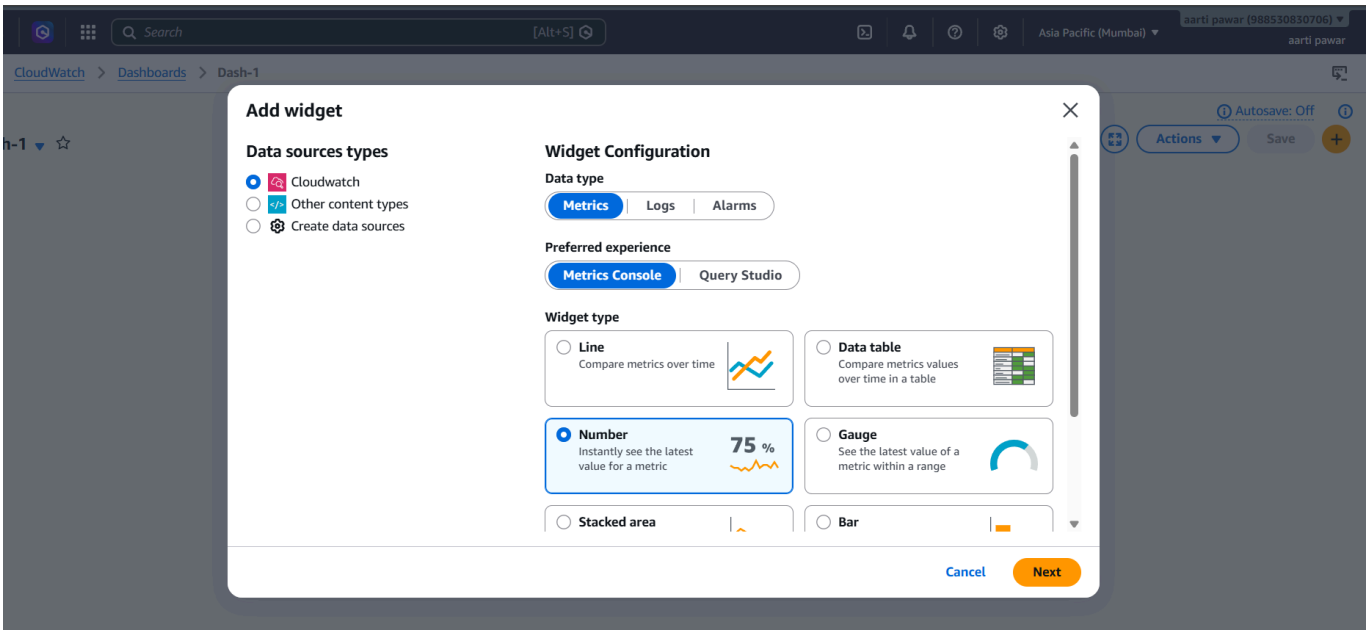


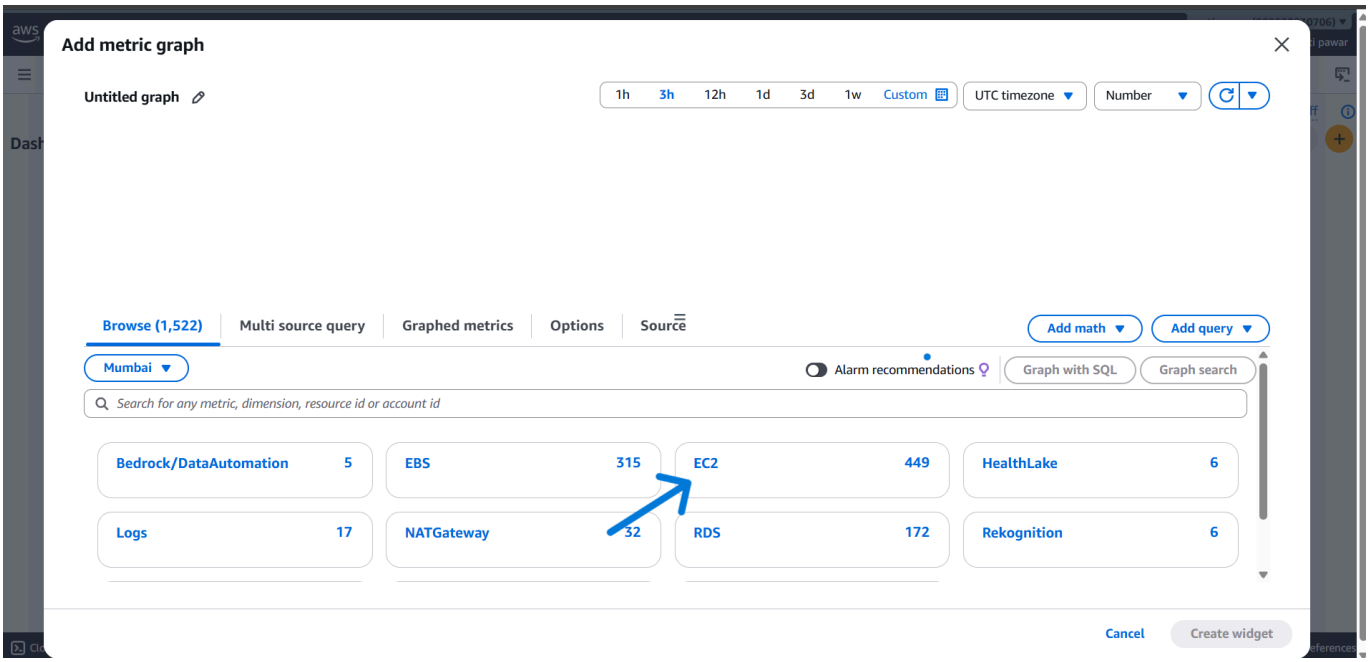


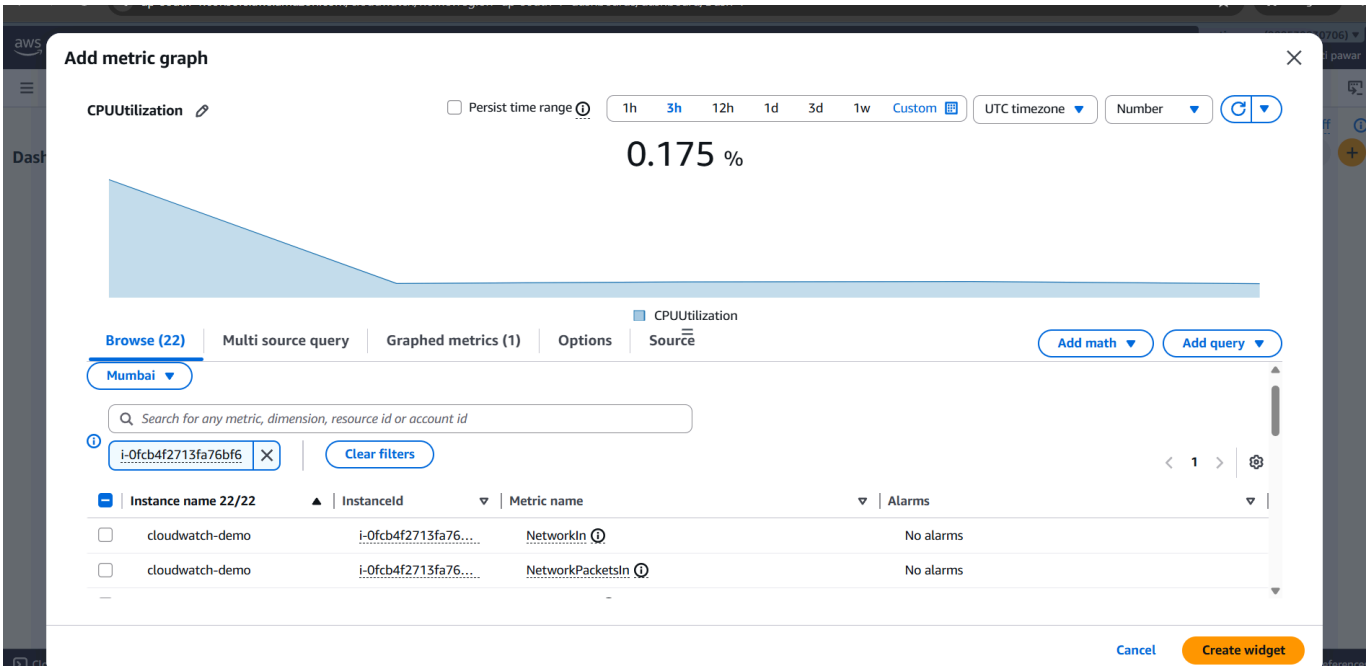


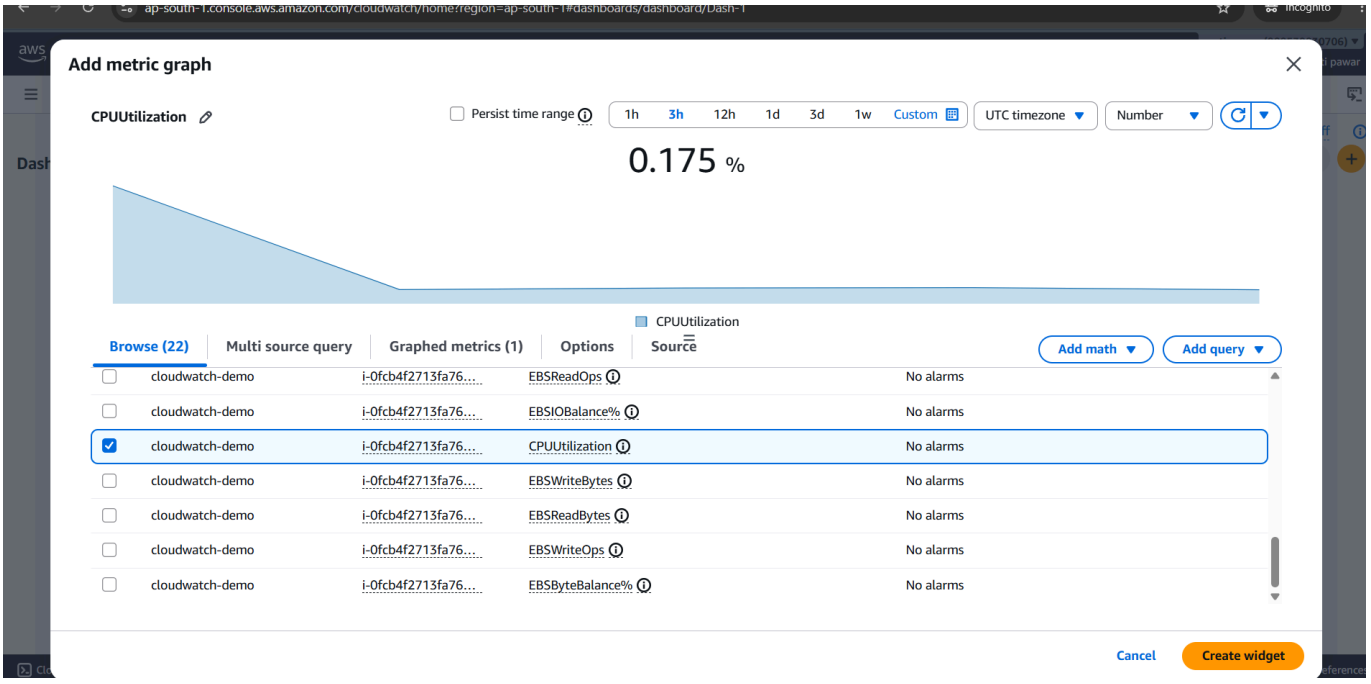
Create dashboard & Share your dashboard publically after you get the link copy that and paste it into browser

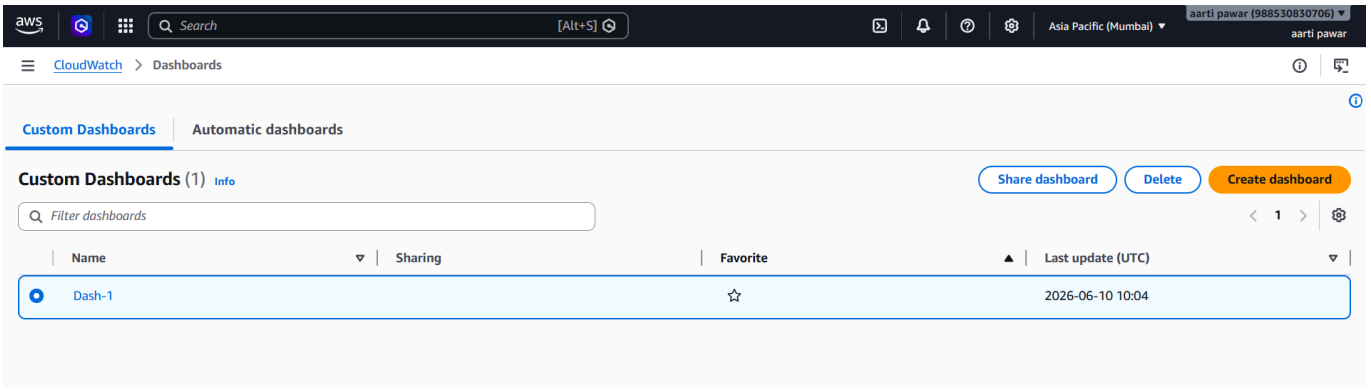


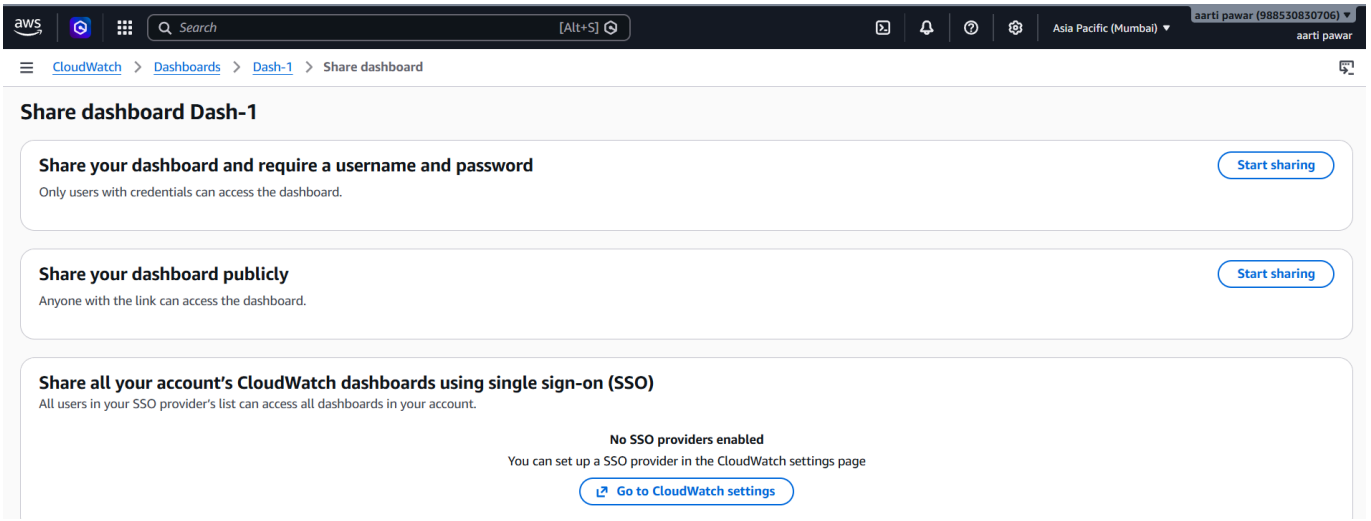


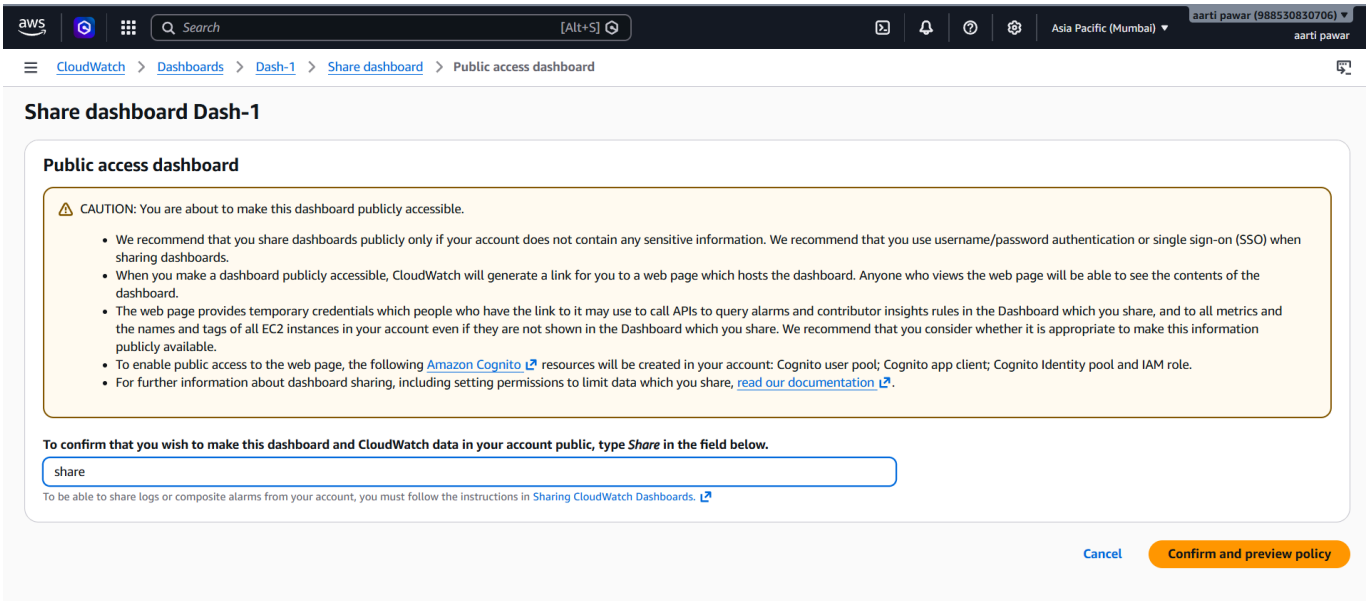

Add metric graph
Untitled graph
1h 3h 12h 1d 3d 1w Custom
UTC timezone
Number
Browse (1,522)
Multi source query
Graphed metrics
Options
Source
Add math
Add query
Mumbai
Alarm recommendations
Graph with SQL
Graph search
Search for any metric, dimension, resource id or account id
Bedrock/DataAutomation 5
EBS 315
EC2 449
HealthLake 6
Logs 17
NATGateway 32
RDS 172
Rekognition 6
Cancel
Create widget

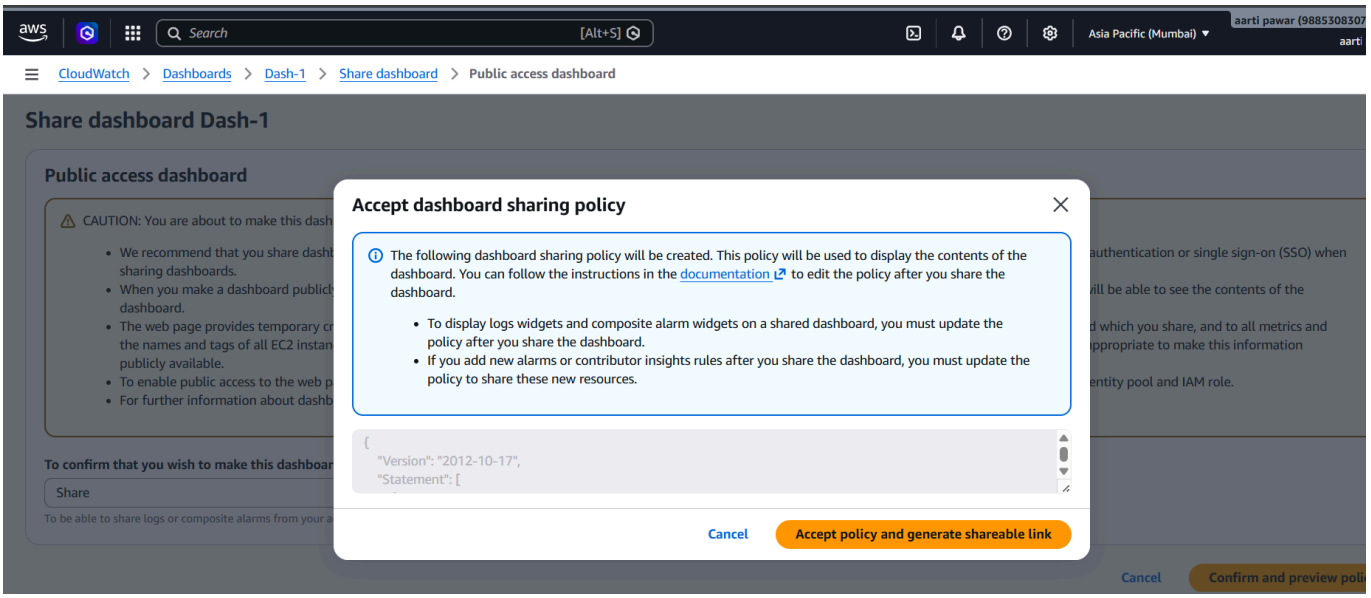

Add metric graph
CPUUtilization
Persist time range
1h 3h 12h 1d 3d 1w Custom
UTC timezone
Number
0.175 %
CPUUtilization
Browse (22)
Multi source query
Graphed metrics (1)
Options
Source
Add math
Add query
Mumbai
Search for any metric, dimension, resource id or account id
i-0fcb4f2713fa76bf6
Clear filters
1
Instance name 22/22
InstanceId
Metric name
Alarms
cloudwatch-demo
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NetworkIn
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Cancel
Create widget

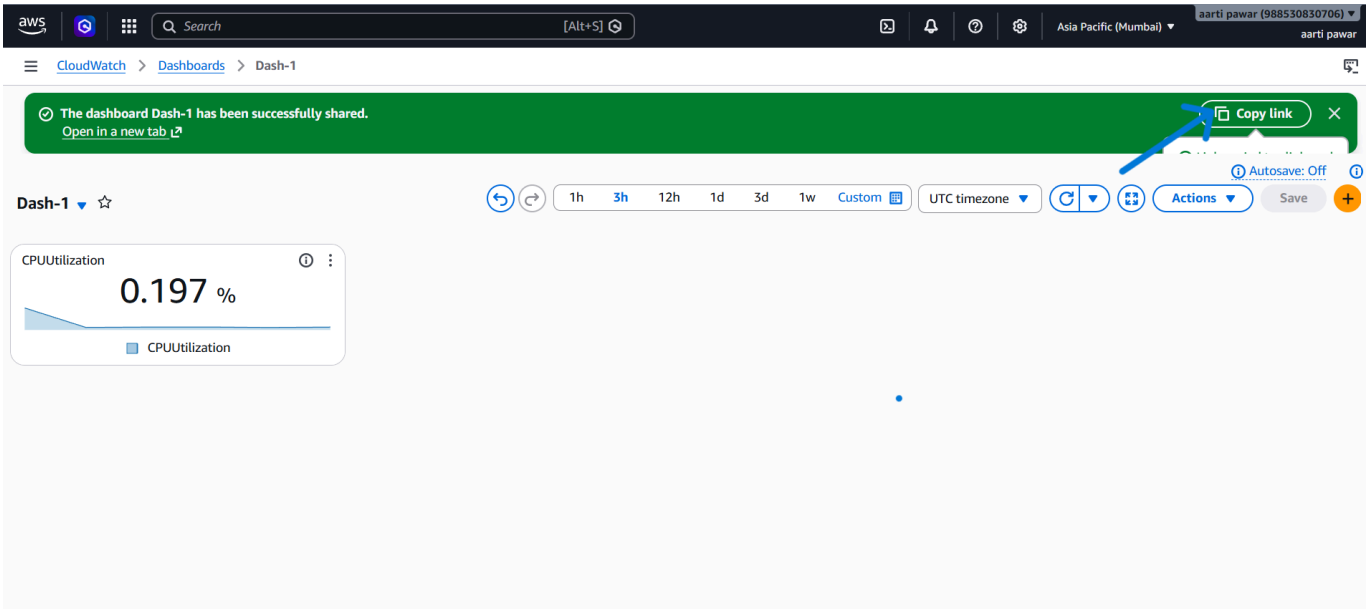

Add metric graph
CPUUtilization
Persist time range
1h 3h 12h 1d 3d 1w Custom
UTC timezone
Number
0.175 %
CPUUtilization
Browse (22)
Multi source query
Graphed metrics (1)
Options
Source
Add math
Add query
cloudwatch-demo i-0fcb4f2713fa76... EBSReadOps No alarms
cloudwatch-demo i-0fcb4f2713fa76... EBSIOBalance% No alarms
cloudwatch-demo i-0fcb4f2713fa76... CPUUtilization No alarms
cloudwatch-demo i-0fcb4f2713fa76... EBSWriteBytes No alarms
cloudwatch-demo i-0fcb4f2713fa76... EBSReadBytes No alarms
cloudwatch-demo i-0fcb4f2713fa76... EBSWriteOps No alarms
cloudwatch-demo i-0fcb4f2713fa76... EBSByteBalance% No alarms
Cancel
Create widget

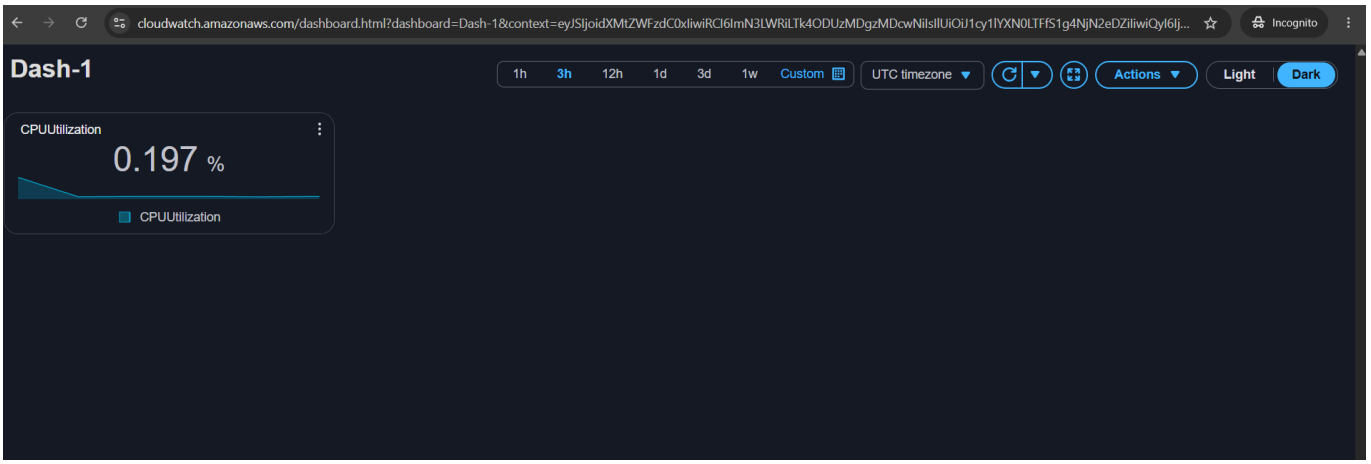

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CloudWatch > Dashboards
Custom Dashboards
Automatic dashboards
Custom Dashboards (1) Info
Share dashboard
Delete
Create dashboard
Filter dashboards
Name
Sharing
Favorite
Last update (UTC)
Dash-1
2026-06-10 10:04


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CloudWatch > Dashboards > Dash-1 > Share dashboard
Share dashboard Dash-1
Share your dashboard and require a username and password
Only users with credentials can access the dashboard.
Start sharing
Share your dashboard publicly
Anyone with the link can access the dashboard.
Start sharing
Share all your account's CloudWatch dashboards using single sign-on (SSO)
All users in your SSO provider's list can access all dashboards in your account.
No SSO providers enabled
You can set up a SSO provider in the CloudWatch settings page
Go to CloudWatch settings


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CloudWatch > Dashboards > Dash-1 > Share dashboard > Public access dashboard
Share dashboard Dash-1
Public access dashboard
CAUTION: You are about to make this dashboard publicly accessible.
We recommend that you share dashboards publicly only if your account does not contain any sensitive information. We recommend that you use username/password authentication or single sign-on (SSO) when sharing dashboards.
When you make a dashboard publicly accessible, CloudWatch will generate a link for you to a web page which hosts the dashboard. Anyone who views the web page will be able to see the contents of the dashboard.
The web page provides temporary credentials which people who have the link to it may use to call APIs to query alarms and contributor insights rules in the Dashboard which you share, and to all metrics and the names and tags of all EC2 instances in your account even if they are not shown in the Dashboard which you share. We recommend that you consider whether it is appropriate to make this information publicly available.
To enable public access to the web page, the following Amazon Cognito resources will be created in your account: Cognito user pool; Cognito app client; Cognito Identity pool and IAM role.
For further information about dashboard sharing, including setting permissions to limit data which you share, read our documentation.
To confirm that you wish to make this dashboard and CloudWatch data in your account public, type Share in the field below.
share
To be able to share logs or composite alarms from your account, you must follow the instructions in Sharing CloudWatch Dashboards.
Cancel
Confirm and preview policy

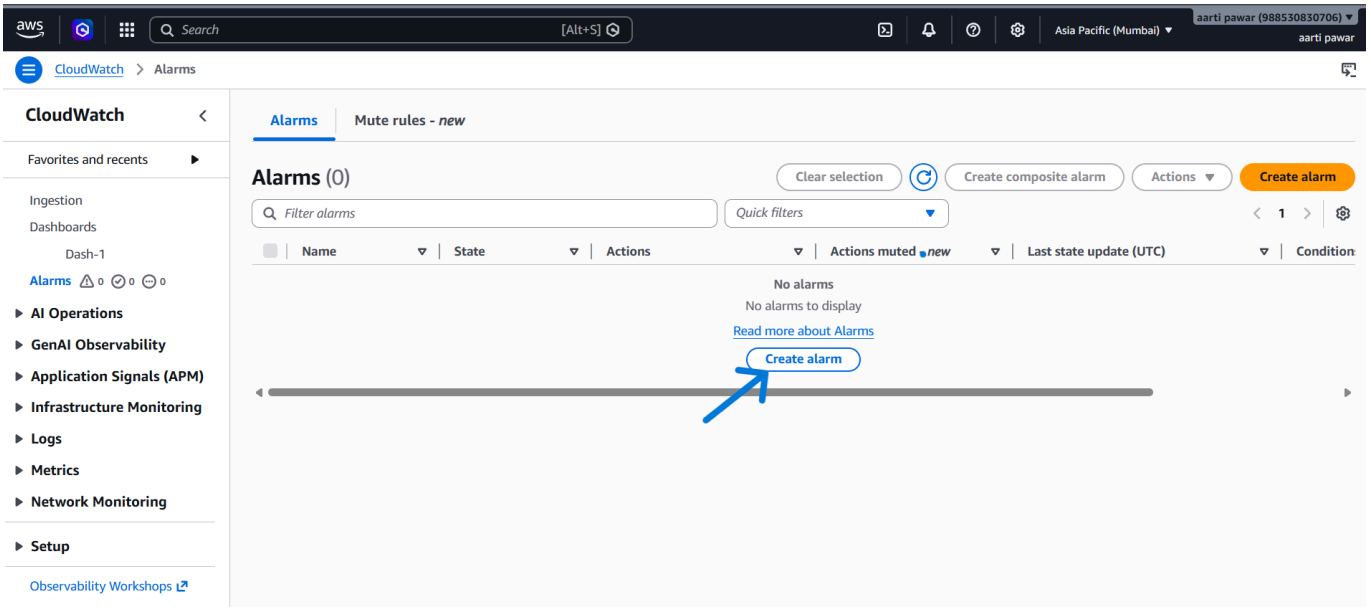

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CloudWatch > Dashboards > Dash-1 > Share dashboard > Public access dashboard
Share dashboard Dash-1
Public access dashboard
Accept dashboard sharing policy
The following dashboard sharing policy will be created. This policy will be used to display the contents of the dashboard. You can follow the instructions in the documentation to edit the policy after you share the dashboard.
To display logs widgets and composite alarm widgets on a shared dashboard, you must update the policy after you share the dashboard.
If you add new alarms or contributor insights rules after you share the dashboard, you must update the policy to share these new resources.
{
"Version": "2012-10-17",
"Statement": [
Cancel
Accept policy and generate shareable link

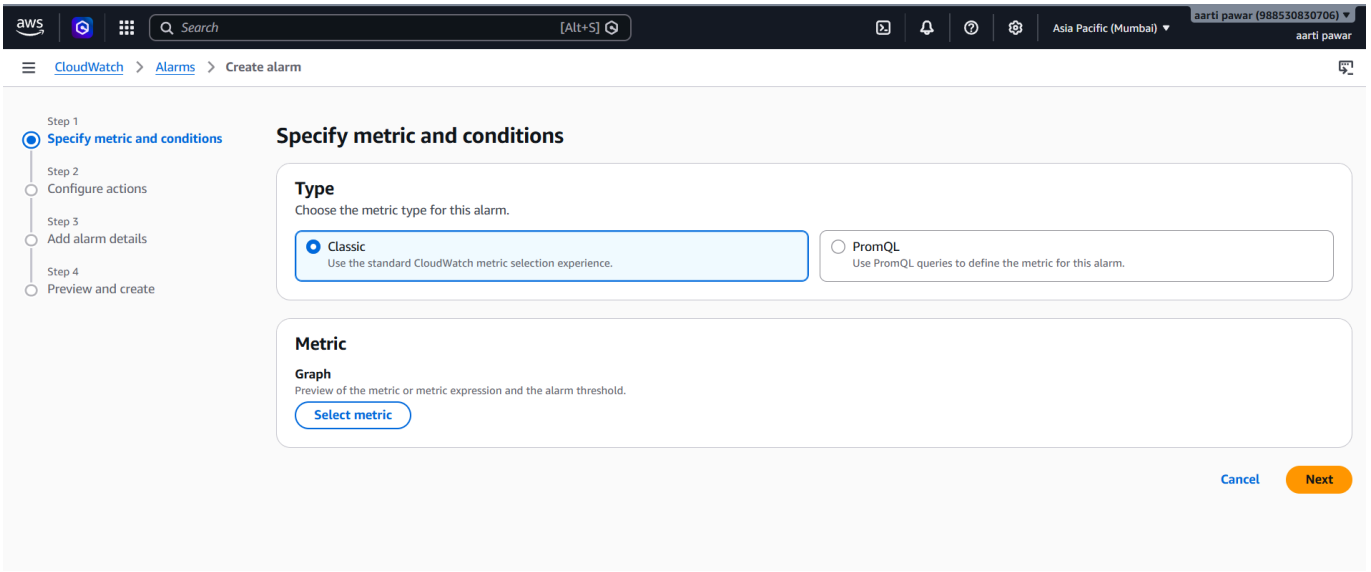

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CloudWatch > Dashboards > Dash-1
The dashboard Dash-1 has been successfully shared.
Open in a new tab
Copy link
Dash-1
1h 3h 12h 1d 3d 1w Custom
UTC timezone
Autosave: Off
Actions
Save
CPUUtilization
0.197 %
CPUUtilization

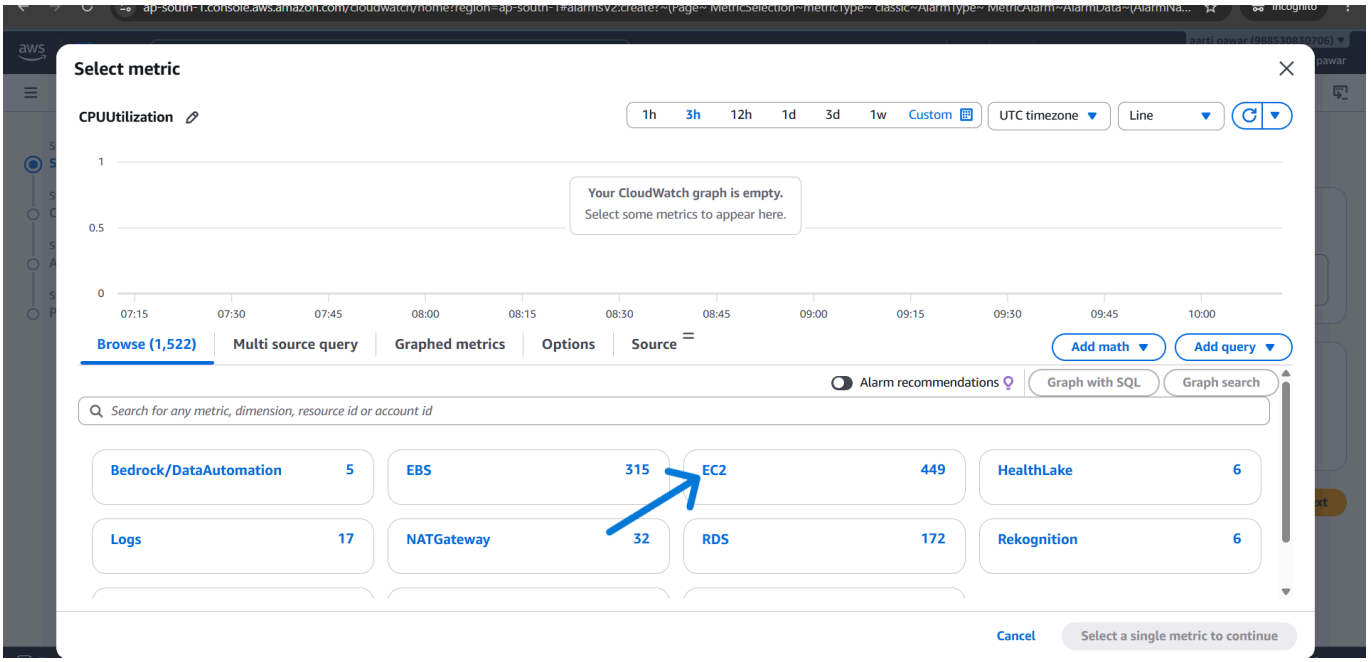


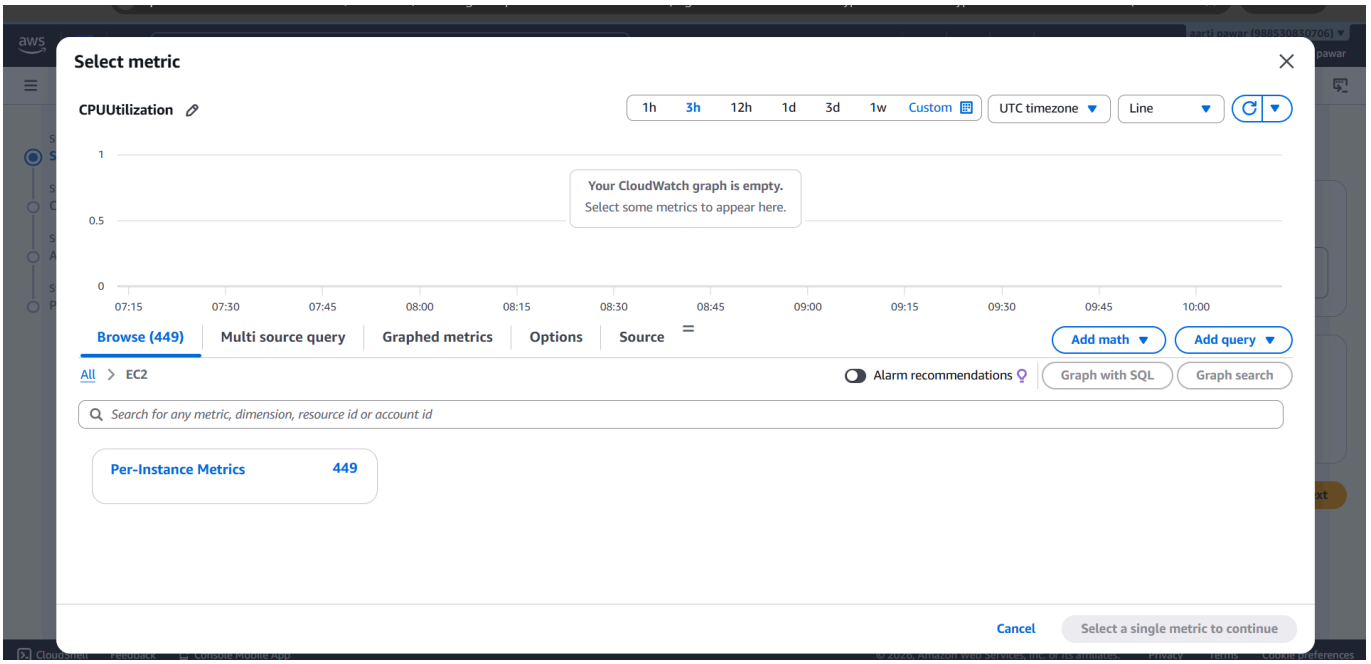
Create Alarm and configure alarm settings

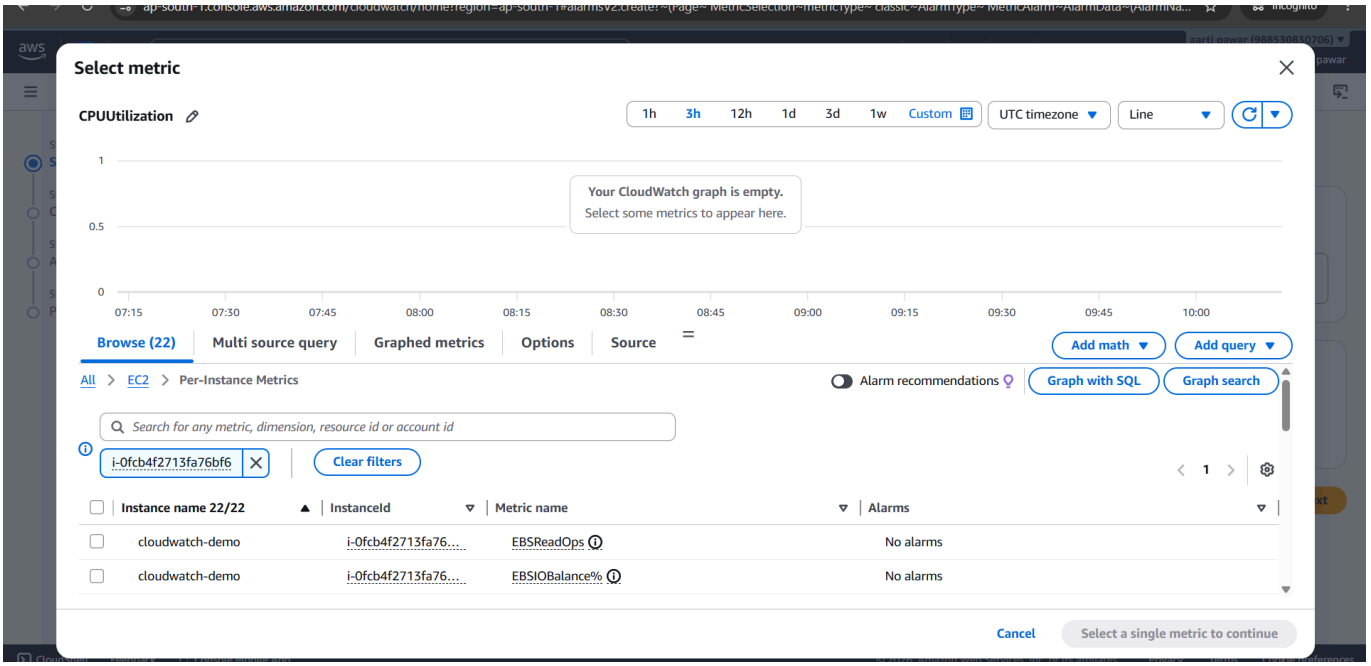
- when your alarm is triggerd then your EC2 instance is stop

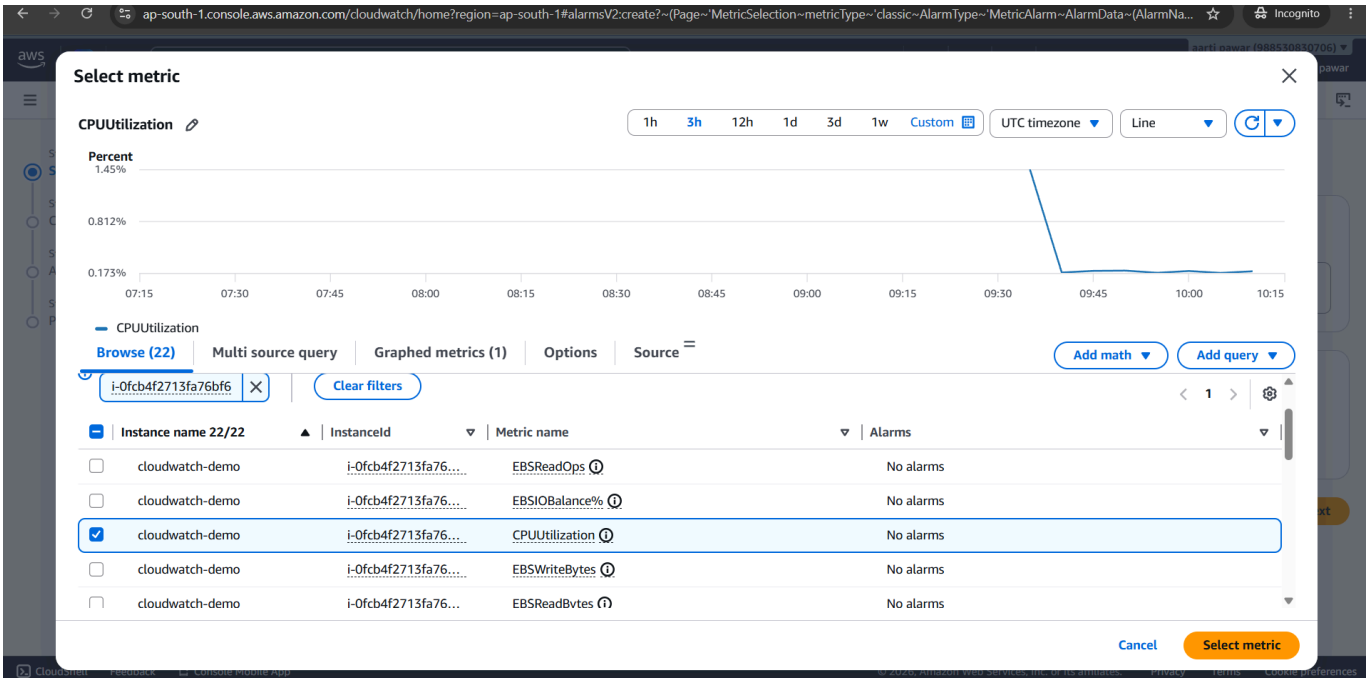


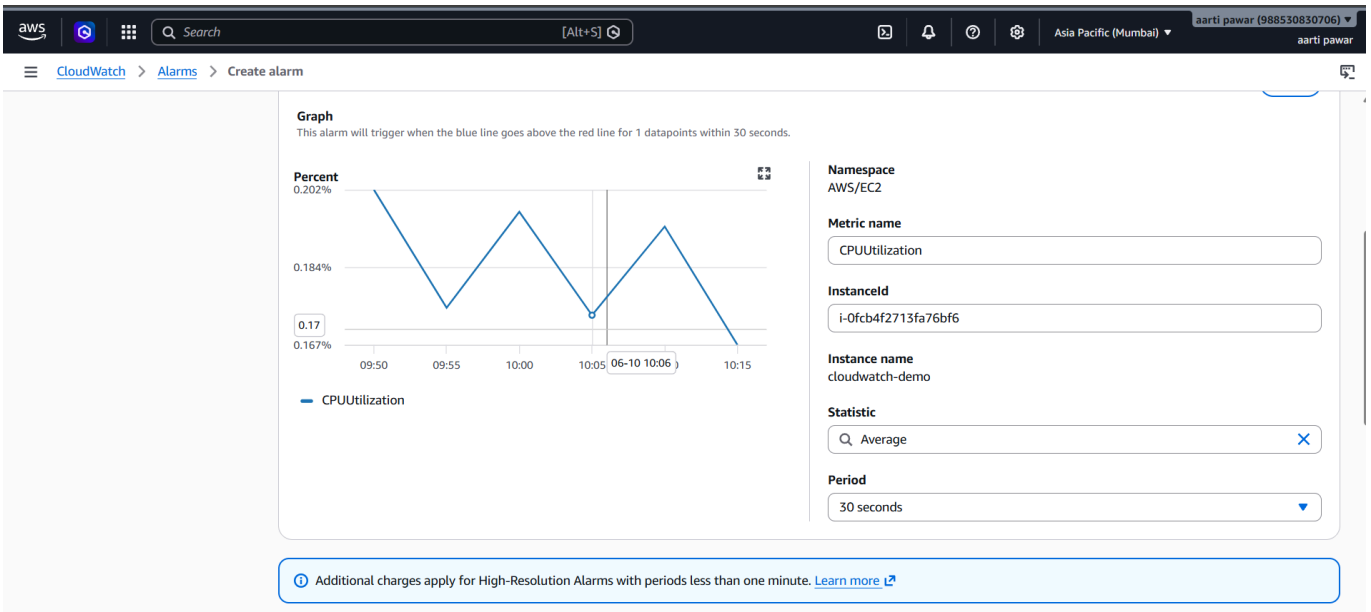


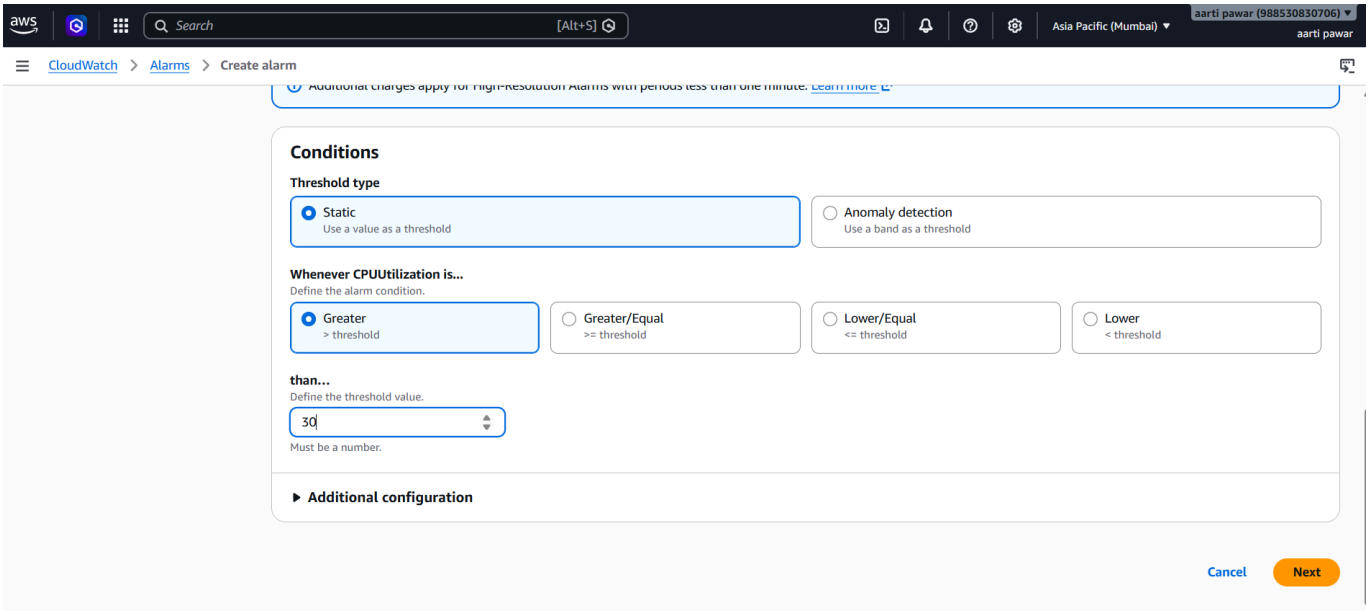

Select metric
CPUUtilization
1h 3h 12h 1d 3d 1w Custom
UTC timezone
Line
Your CloudWatch graph is empty.
Select some metrics to appear here.
Browse (1,522)
Multi source query
Graphed metrics
Options
Source
Add math
Add query
Alarm recommendations
Graph with SQL
Graph search
Search for any metric, dimension, resource id or account id
Bedrock/DataAutomation 5
EBS 315
EC2 449
HealthLake 6
Logs 17
NATGateway 32
RDS 172
Rekognition 6
Cancel
Select a single metric to continue

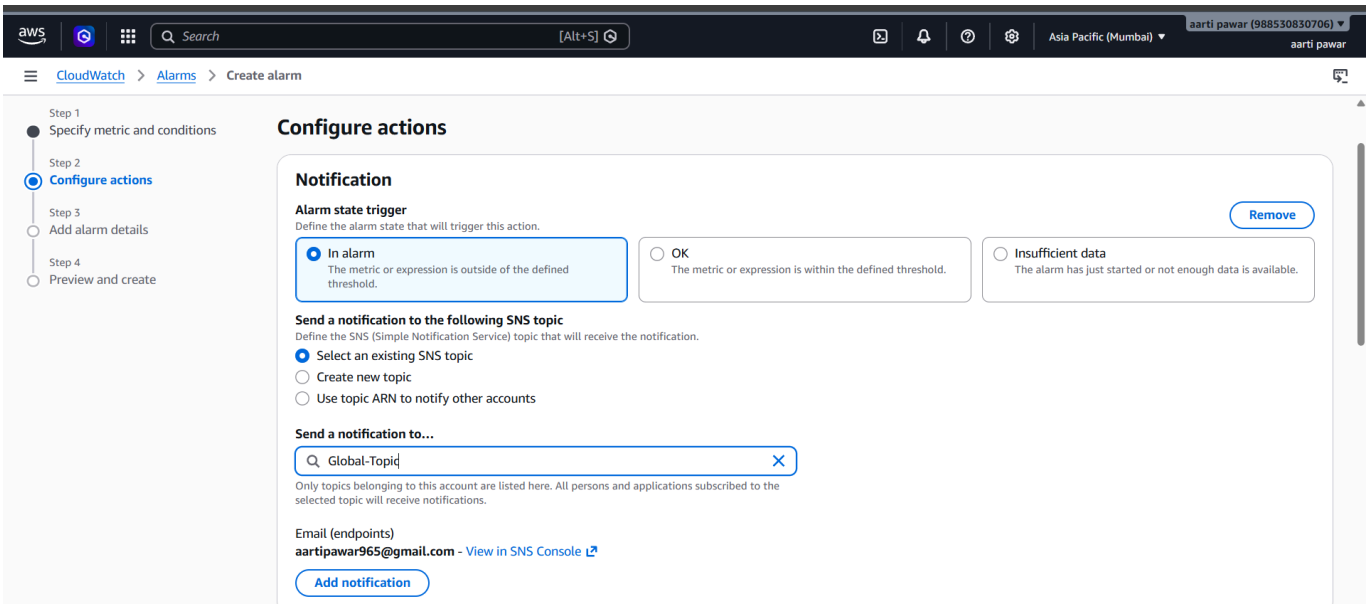

Select metric
CPUUtilization
1h 3h 12h 1d 3d 1w Custom
UTC timezone
Line
Your CloudWatch graph is empty.
Select some metrics to appear here.
Browse (449)
Multi source query
Graphed metrics
Options
Source
Add math
Add query
All > EC2
Alarm recommendations
Graph with SQL
Graph search
Search for any metric, dimension, resource id or account id
Per-Instance Metrics 449
Cancel
Select a single metric to continue

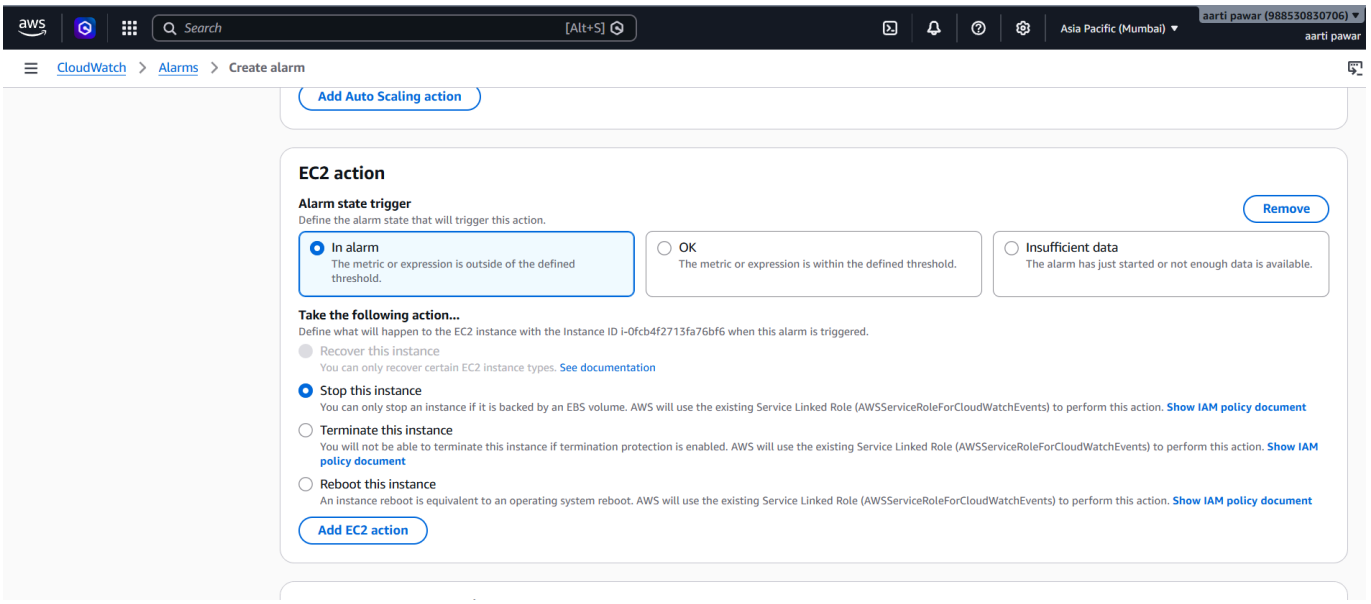

Select metric
CPUUtilization
1h 3h 12h 1d 3d 1w Custom
UTC timezone
Line
Your CloudWatch graph is empty.
Select some metrics to appear here.
Browse (22)
Multi source query
Graphed metrics
Options
Source
Add math
Add query
All > EC2 > Per-Instance Metrics
Alarm recommendations
Graph with SQL
Graph search
Search for any metric, dimension, resource id or account id
i-0fcb4f2713fa76bf6
Clear filters
Instance name 22/22
InstanceId
Metric name
Alarms
cloudwatch-demo i-0fcb4f2713fa76... EBSReadOps No alarms
cloudwatch-demo i-0fcb4f2713fa76... EBSIOBalance% No alarms
Cancel
Select a single metric to continue

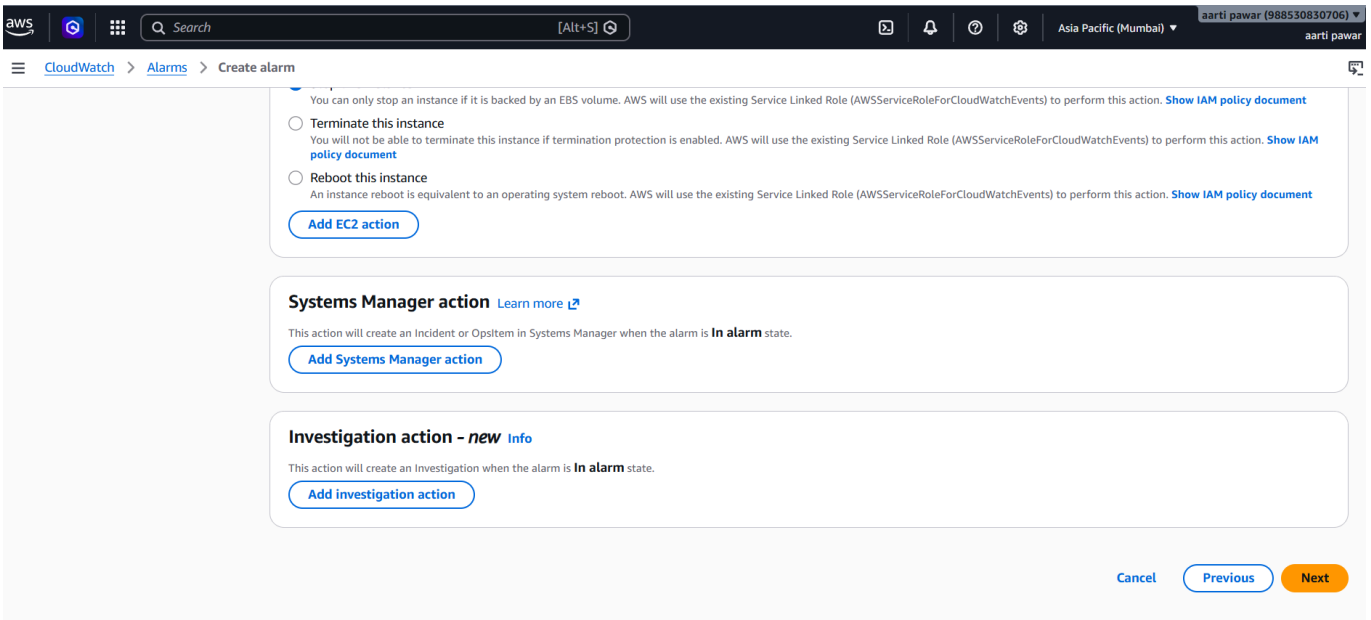

Select metric
CPUUtilization
1h 3h 12h 1d 3d 1w Custom
UTC timezone
Line
Percent
1.45%
0.812%
0.173%
07:15 07:30 07:45 08:00 08:15 08:30 08:45 09:00 09:15 09:30 09:45 10:00 10:15
CPUUtilization
Browse (22)
Multi source query
Graphed metrics (1)
Options
Source
Add math
Add query
i-0fcb4f2713fa76bf6
Clear filters
Instance name 22/22
InstanceId
Metric name
Alarms
cloudwatch-demo i-0fcb4f2713fa76... EBSReadOps No alarms
cloudwatch-demo i-0fcb4f2713fa76... EBSIOBalance% No alarms
cloudwatch-demo i-0fcb4f2713fa76... CPUUtilization No alarms
cloudwatch-demo i-0fcb4f2713fa76... EBSWriteBytes No alarms
cloudwatch-demo i-0fcb4f2713fa76... EBSReadBytes No alarms
Cancel
Select metric

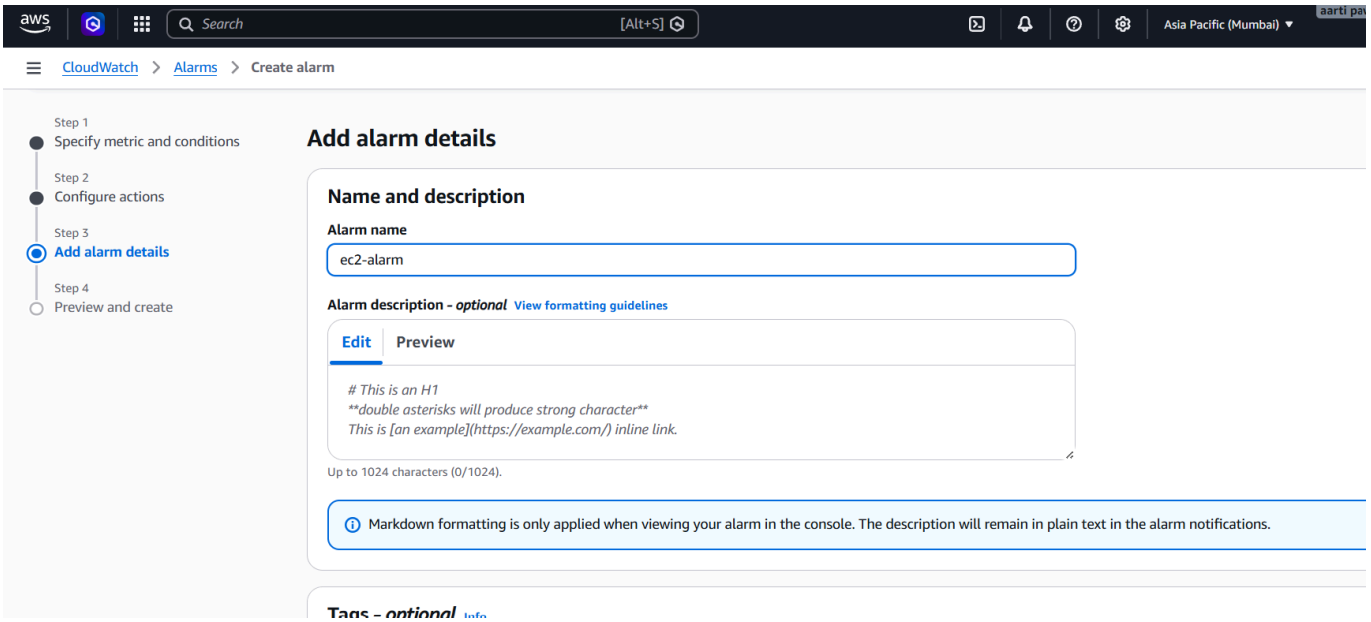

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Asia Pacific (Mumbai)
aarti pawar (988530830706)
aarti pawar
CloudWatch > Alarms > Create alarm
Graph
This alarm will trigger when the blue line goes above the red line for 1 datapoints within 30 seconds.
Percent
0.202%
0.184%
0.17
0.167%
09:50 09:55 10:00 10:05 06-10 10:06 10:15
CPUUtilization
Namespace
AWS/EC2
Metric name
CPUUtilization
InstanceId
i-0fcb4f2713fa76bf6
Instance name
cloudwatch-demo
Statistic
Average
Period
30 seconds
Additional charges apply for High-Resolution Alarms with periods less than one minute. Learn more

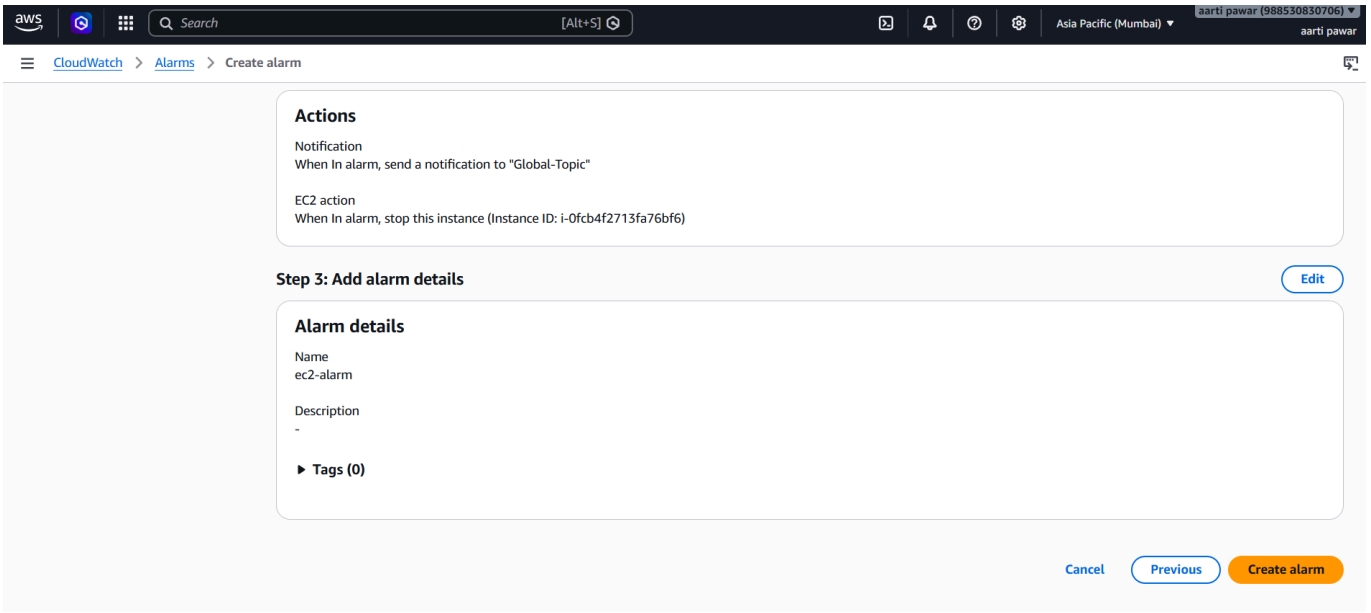

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CloudWatch > Alarms > Create alarm
Conditions
Threshold type
Static
Use a value as a threshold
Anomaly detection
Use a band as a threshold
Whenever CPUUtilization is...
Define the alarm condition.
Greater
> threshold
Greater/Equal
>= threshold
Lower/Equal
<= threshold
Lower
< threshold
than...
Define the threshold value.
30
Must be a number.
Additional configuration
Cancel
Next

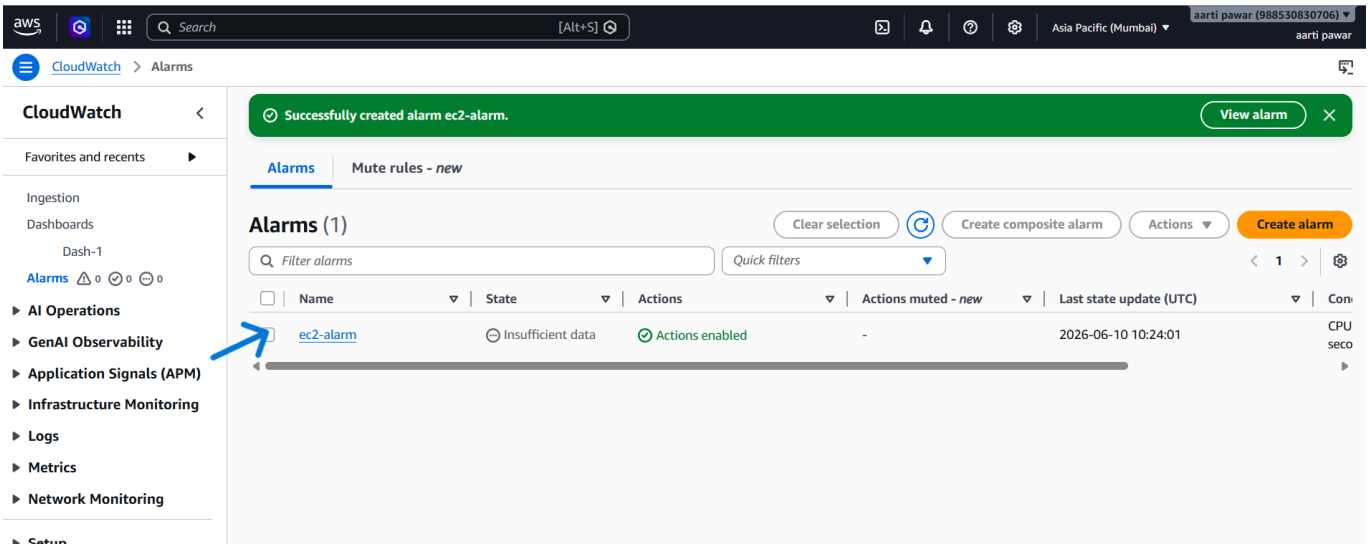

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CloudWatch > Alarms > Create alarm
Step 1
Specify metric and conditions
Step 2
Configure actions
Step 3
Add alarm details
Step 4
Preview and create
Configure actions
Notification
Alarm state trigger
Define the alarm state that will trigger this action.
Remove
In alarm
The metric or expression is outside of the defined threshold.
OK
The metric or expression is within the defined threshold.
Insufficient data
The alarm has just started or not enough data is available.
Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.
Select an existing SNS topic
Create new topic
Use topic ARN to notify other accounts
Send a notification to...
Global-Topic
Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.
Email (endpoints)
aartipawar965@gmail.com - View in SNS Console
Add notification


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CloudWatch > Alarms > Create alarm
Add Auto Scaling action
EC2 action
Alarm state trigger
Define the alarm state that will trigger this action.
Remove
In alarm
The metric or expression is outside of the defined threshold.
OK
The metric or expression is within the defined threshold.
Insufficient data
The alarm has just started or not enough data is available.
Take the following action...
Define what will happen to the EC2 instance with the Instance ID i-0fcb4f2713fa76bf6 when this alarm is triggered.
Recover this instance
You can only recover certain EC2 instance types. See documentation
Stop this instance
You can only stop an instance if it is backed by an EBS volume. AWS will use the existing Service Linked Role (AWSServiceRoleForCloudWatchEvents) to perform this action. Show IAM policy document
Terminate this instance
You will not be able to terminate this instance if termination protection is enabled. AWS will use the existing Service Linked Role (AWSServiceRoleForCloudWatchEvents) to perform this action. Show IAM policy document
Reboot this instance
An instance reboot is equivalent to an operating system reboot. AWS will use the existing Service Linked Role (AWSServiceRoleForCloudWatchEvents) to perform this action. Show IAM policy document
Add EC2 action


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aarti pawar
CloudWatch > Alarms > Create alarm
You can only stop an instance if it is backed by an EBS volume. AWS will use the existing Service Linked Role (AWSServiceRoleForCloudWatchEvents) to perform this action. Show IAM policy document
Terminate this instance
You will not be able to terminate this instance if termination protection is enabled. AWS will use the existing Service Linked Role (AWSServiceRoleForCloudWatchEvents) to perform this action. Show IAM policy document
Reboot this instance
An instance reboot is equivalent to an operating system reboot. AWS will use the existing Service Linked Role (AWSServiceRoleForCloudWatchEvents) to perform this action. Show IAM policy document
Add EC2 action
Systems Manager action Learn more
This action will create an Incident or OpsItem in Systems Manager when the alarm is In alarm state.
Add Systems Manager action
Investigation action - new Info
This action will create an Investigation when the alarm is In alarm state.
Add investigation action
Cancel
Previous
Next

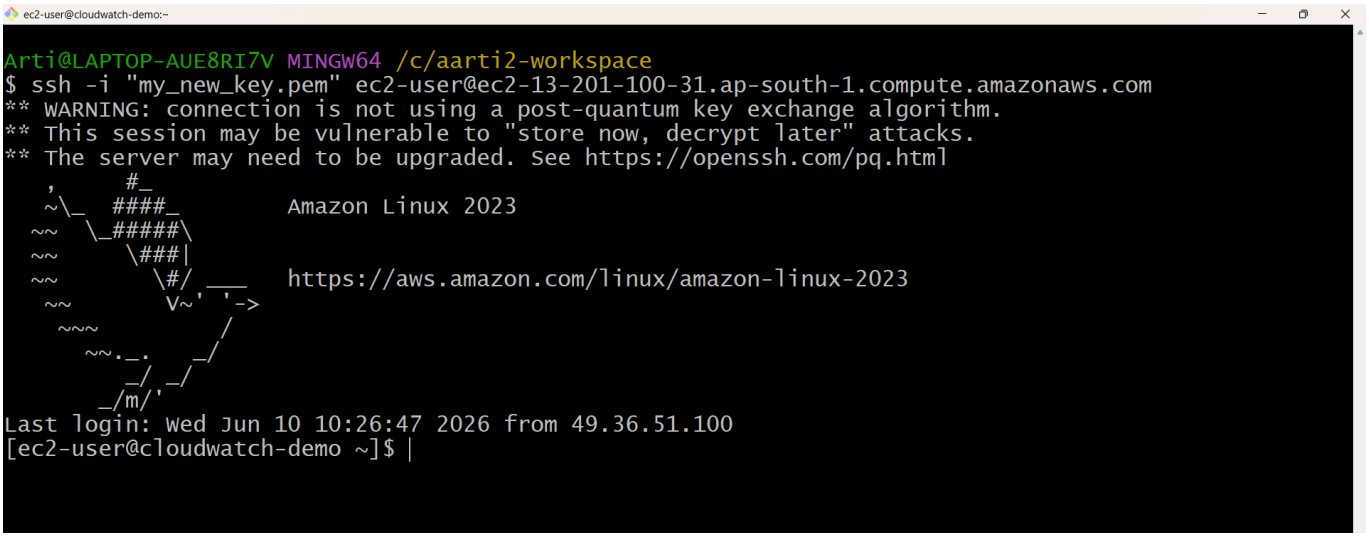


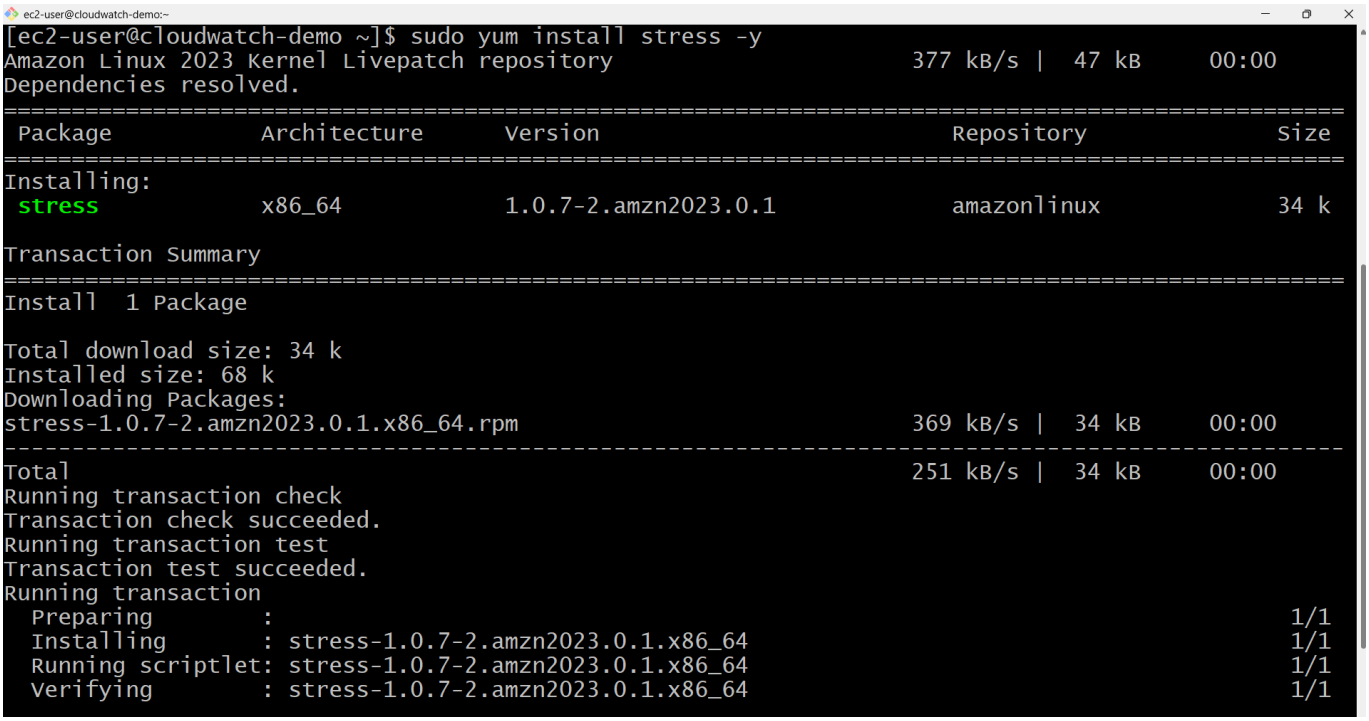


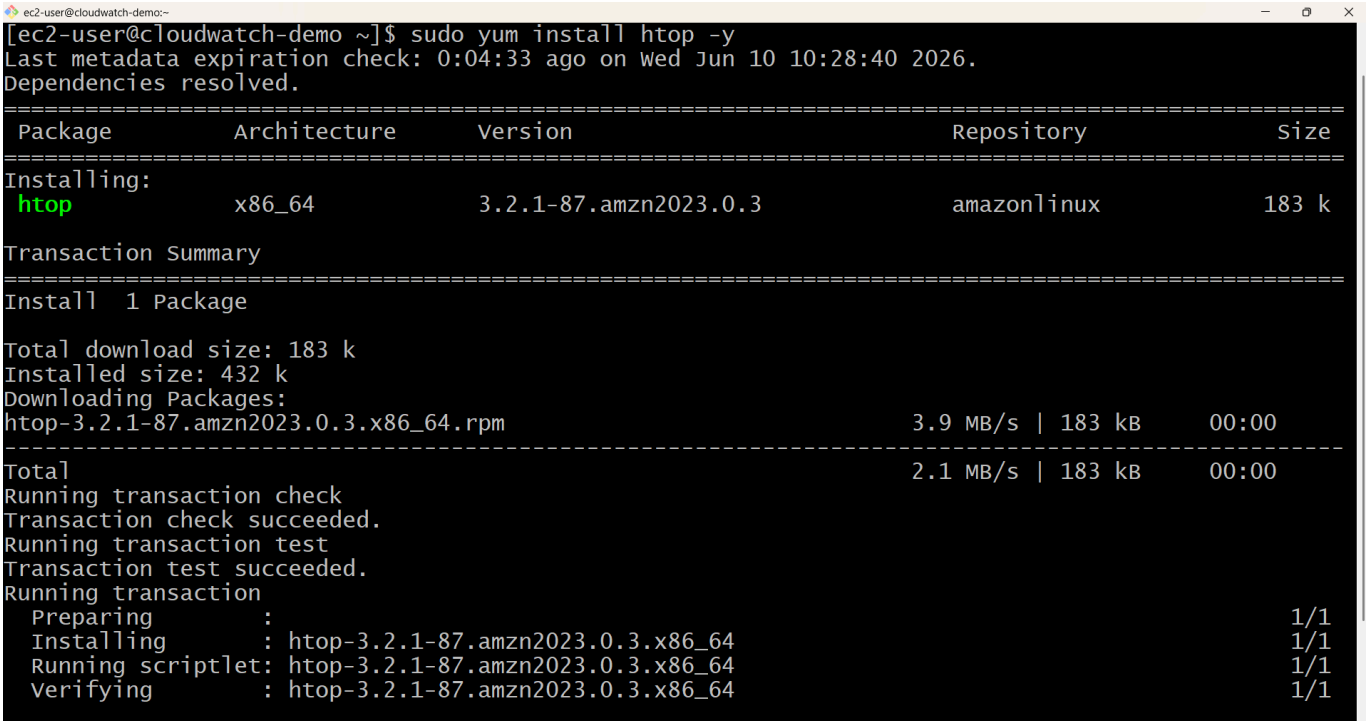


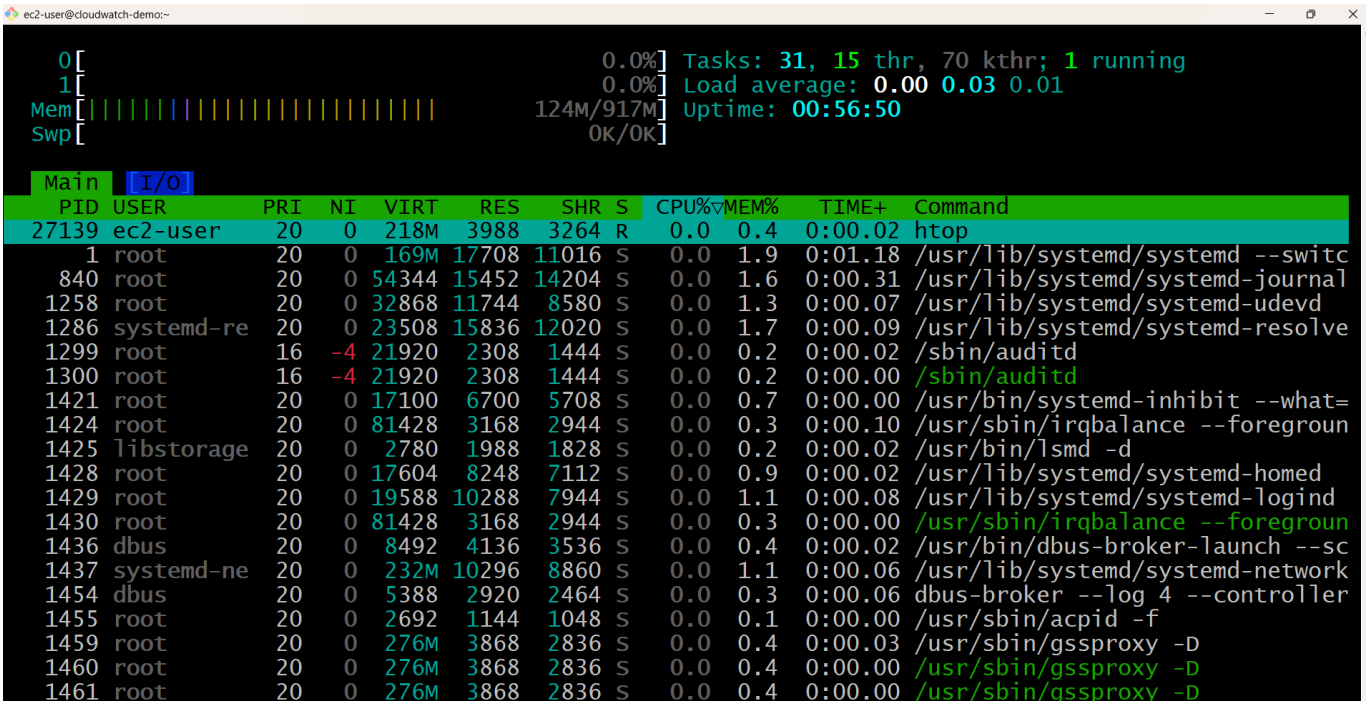
Connect to your EC2 instance

```
sudo yum install stress -y
sudo yum install htop -y
htop
stress --cpu 80 --io 4 --vm 2 --vm-bytes 128M --timeout 10m &
stress: info: [27190] dispatching hogs: 80 cpu, 4 io, 2 vm, 0 hdd
jobs
```






ec2-user@cloudwatch-demo:~
[ec2-user@cloudwatch-demo ~]$ sudo yum install htop -y
Last metadata expiration check: 0:04:33 ago on Wed Jun 10 10:28:40 2026.
Dependencies resolved.
Package Architecture Version Repository Size
Installing:
htop x86_64 3.2.1-87.amzn2023.0.3 amazonlinux 183 k
Transaction Summary
Install 1 Package
Total download size: 183 k
Installed size: 432 k
Downloading Packages:
htop-3.2.1-87.amzn2023.0.3.x86_64.rpm 3.9 MB/s | 183 kB 00:00
Total 2.1 MB/s | 183 kB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing :
Installing : htop-3.2.1-87.amzn2023.0.3.x86_64
Running scriptlet: htop-3.2.1-87.amzn2023.0.3.x86_64
Verifying : htop-3.2.1-87.amzn2023.0.3.x86_64


ec2-user@cloudwatch-demo:~
0[0.0%] Tasks: 31, 15 thr, 70 kthr; 1 running
1[0.0%] Load average: 0.00 0.03 0.01
Mem[124M/917M] Uptime: 00:56:50
Swp[0K/0K]
Main I/O
PID USER PRI NI VIRT RES SHR S CPU%▽MEM% TIME+ Command
27139 ec2-user 20 0 218M 3988 3264 R 0.0 0.4 0:00.02 htop
1 root 20 0 169M 17708 11016 S 0.0 1.9 0:01.18 /usr/lib/systemd/systemd --switc
840 root 20 0 54344 15452 14204 S 0.0 1.6 0:00.31 /usr/lib/systemd/systemd-journal
1258 root 20 0 32868 11744 8580 S 0.0 1.3 0:00.07 /usr/lib/systemd/systemd-udevd
1286 systemd-re 20 0 23508 15836 12020 S 0.0 1.7 0:00.09 /usr/lib/systemd/systemd-resolve
1299 root 16 -4 21920 2308 1444 S 0.0 0.2 0:00.02 /sbin/auditd
1300 root 16 -4 21920 2308 1444 S 0.0 0.2 0:00.00 /sbin/auditd
1421 root 20 0 17100 6700 5708 S 0.0 0.7 0:00.00 /usr/bin/systemd-inhibit --what=
1424 root 20 0 81428 3168 2944 S 0.0 0.3 0:00.10 /usr/sbin/irqbalance --foregroun
1425 libstorage 20 0 2780 1988 1828 S 0.0 0.2 0:00.02 /usr/bin/lsmd -d
1428 root 20 0 17604 8248 7112 S 0.0 0.9 0:00.02 /usr/lib/systemd/systemd-homed
1429 root 20 0 19588 10288 7944 S 0.0 1.1 0:00.08 /usr/lib/systemd/systemd-logind
1430 root 20 0 81428 3168 2944 S 0.0 0.3 0:00.00 /usr/sbin/irqbalance --foregroun
1436 dbus 20 0 8492 4136 3536 S 0.0 0.4 0:00.02 /usr/bin/dbus-broker-launch --sc
1437 systemd-ne 20 0 232M 10296 8860 S 0.0 1.1 0:00.06 /usr/lib/systemd/systemd-network
1454 dbus 20 0 5388 2920 2464 S 0.0 0.3 0:00.06 dbus-broker --log 4 --controller
1455 root 20 0 2692 1144 1048 S 0.0 0.1 0:00.00 /usr/sbin/acpid -f
1459 root 20 0 276M 3868 2836 S 0.0 0.4 0:00.03 /usr/sbin/gssproxy -D
1460 root 20 0 276M 3868 2836 S 0.0 0.4 0:00.00 /usr/sbin/gssproxy -D
1461 root 20 0 276M 3868 2836 S 0.0 0.4 0:00.00 /usr/sbin/gssproxy -D

```
ec2-user@cloudwatch-demo:~
[ec2-user@cloudwatch-demo ~]$ htop
[ec2-user@cloudwatch-demo ~]$ stress --help
`stress' imposes certain types of compute stress on your system

Usage: stress [OPTION [ARG]] ...
 -?, --help         show this help statement
     --version      show version statement
 -v, --verbose      be verbose
 -q, --quiet        be quiet
 -n, --dry-run      show what would have been done
 -t, --timeout N    timeout after N seconds
     --backoff N    wait factor of N microseconds before work starts
 -c, --cpu N        spawn N workers spinning on sqrt()
 -i, --io N         spawn N workers spinning on sync()
 -m, --vm N         spawn N workers spinning on malloc()/free()
     --vm-bytes B   malloc B bytes per vm worker (default is 256MB)
     --vm-stride B  touch a byte every B bytes (default is 4096)
     --vm-hang N    sleep N secs before free (default none, 0 is inf)
     --vm-keep      redirty memory instead of freeing and reallocating
 -d, --hdd N        spawn N workers spinning on write()/unlink()
     --hdd-bytes B  write B bytes per hdd worker (default is 1GB)

Example: stress --cpu 8 --io 4 --vm 2 --vm-bytes 128M --timeout 10s

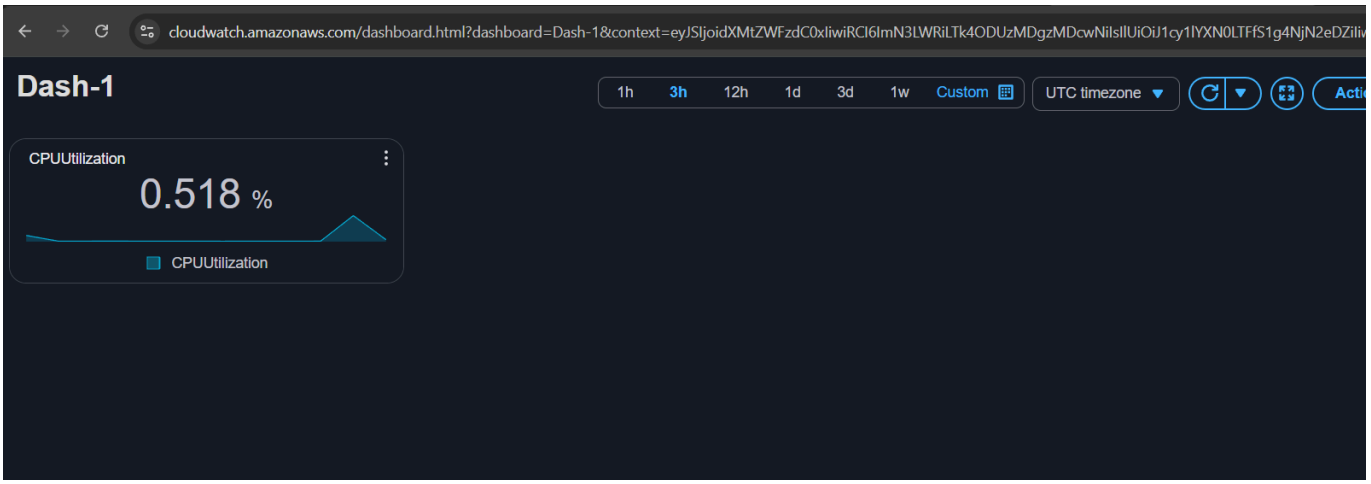
Note: Numbers may be suffixed with s,m,h,d,y (time) or B,K,M,G (size).
[ec2-user@cloudwatch-demo ~]$
```

```
ec2-user@cloudwatch-demo:~
[ec2-user@cloudwatch-demo ~]$ stress --cpu 80 --io 4 --vm 2 --vm-bytes 128M --timeout 10m &
[1] 27190
[ec2-user@cloudwatch-demo ~]$ stress: info: [27190] dispatching hogs: 80 cpu, 4 io, 2 vm, 0 hdd
^C
[ec2-user@cloudwatch-demo ~]$ jobs
[1]+  Running                 stress --cpu 80 --io 4 --vm 2 --vm-bytes 128M --timeout 10m &
[ec2-user@cloudwatch-demo ~]$
```

```
ec2-user@cloudwatch-demo:~
  0[||||||||||||||||||||||||||||||||||100.0%] Tasks: 118, 15 thr, 70 kthr; 2 running
  1[||||||||||||||||||||||||||||||||||100.0%] Load average: 54.41 15.67 5.38
  Mem[||||||||||||||||||||||||||||321M/917M] Uptime: 01:00:08
  Swp[                               0K/0K]

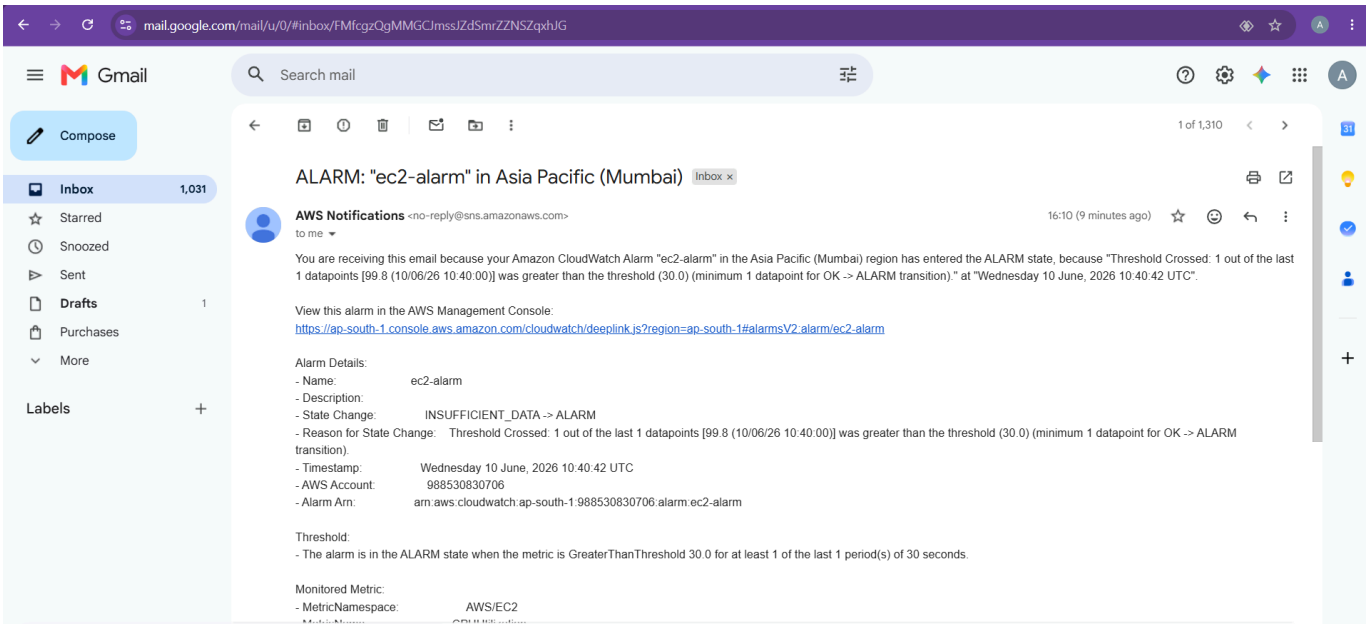
  Main  I/O
    PID USER       PRI  NI  VIRT   RES   SHR S  CPU%▽MEM%   TIME+  Command
  27197 ec2-user    20   0  3536   108     0 R   2.9  0.0  0:01.41 stress --cpu 80 --io 4 --vm 2 --
  27204 ec2-user    20   0  3536   108     0 R   2.9  0.0  0:01.37 stress --cpu 80 --io 4 --vm 2 --
  27209 ec2-user    20   0  3536   108     0 R   2.9  0.0  0:01.37 stress --cpu 80 --io 4 --vm 2 --
  27231 ec2-user    20   0  3536   108     0 R   2.9  0.0  0:01.38 stress --cpu 80 --io 4 --vm 2 --
  27271 ec2-user    20   0  3536   108     0 R   2.9  0.0  0:01.42 stress --cpu 80 --io 4 --vm 2 --
  27272 ec2-user    20   0  3536   108     0 R   2.9  0.0  0:01.38 stress --cpu 80 --io 4 --vm 2 --
  27191 ec2-user    20   0  3536   108     0 R   2.3  0.0  0:01.36 stress --cpu 80 --io 4 --vm 2 --
  27192 ec2-user    20   0  3536   108     0 R   2.3  0.0  0:01.40 stress --cpu 80 --io 4 --vm 2 --
  27193 ec2-user    20   0  131M 64524   268 R   2.3  6.9  0:01.36 stress --cpu 80 --io 4 --vm 2 --
  27194 ec2-user    20   0  3536   108     0 R   2.3  0.0  0:01.38 stress --cpu 80 --io 4 --vm 2 --
  27195 ec2-user    20   0  3536   108     0 R   2.3  0.0  0:01.41 stress --cpu 80 --io 4 --vm 2 --
  27196 ec2-user    20   0  131M 92656   268 R   2.3  9.9  0:01.39 stress --cpu 80 --io 4 --vm 2 --
  27198 ec2-user    20   0  3536   108     0 R   2.3  0.0  0:01.36 stress --cpu 80 --io 4 --vm 2 --
  27199 ec2-user    20   0  3536   108     0 R   2.3  0.0  0:01.36 stress --cpu 80 --io 4 --vm 2 --
  27200 ec2-user    20   0  3536   108     0 R   2.3  0.0  0:01.39 stress --cpu 80 --io 4 --vm 2 --
  27201 ec2-user    20   0  3536   108     0 R   2.3  0.0  0:01.36 stress --cpu 80 --io 4 --vm 2 --
  27202 ec2-user    20   0  3536   108     0 R   2.3  0.0  0:01.36 stress --cpu 80 --io 4 --vm 2 --
  27203 ec2-user    20   0  3536   108     0 R   2.3  0.0  0:01.41 stress --cpu 80 --io 4 --vm 2 --
  27205 ec2-user    20   0  3536   108     0 R   2.3  0.0  0:01.40 stress --cpu 80 --io 4 --vm 2 --
  27206 ec2-user    20   0  3536   108     0 R   2.3  0.0  0:01.37 stress --cpu 80 --io 4 --vm 2 --
F1Help  F2Setup F3Search F4Filter F5Tree  F6SortBy F7Nice -F8Nice +F9Kill  F10Quit
```

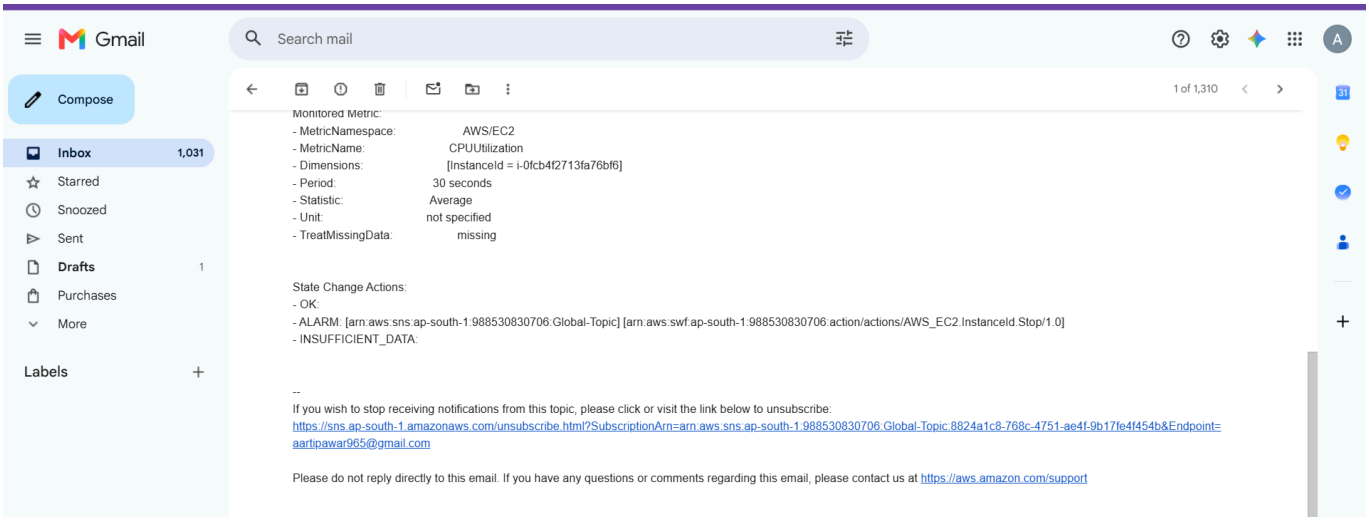
Access your Dashboard and refresh dashboard & you see your instance load was increases



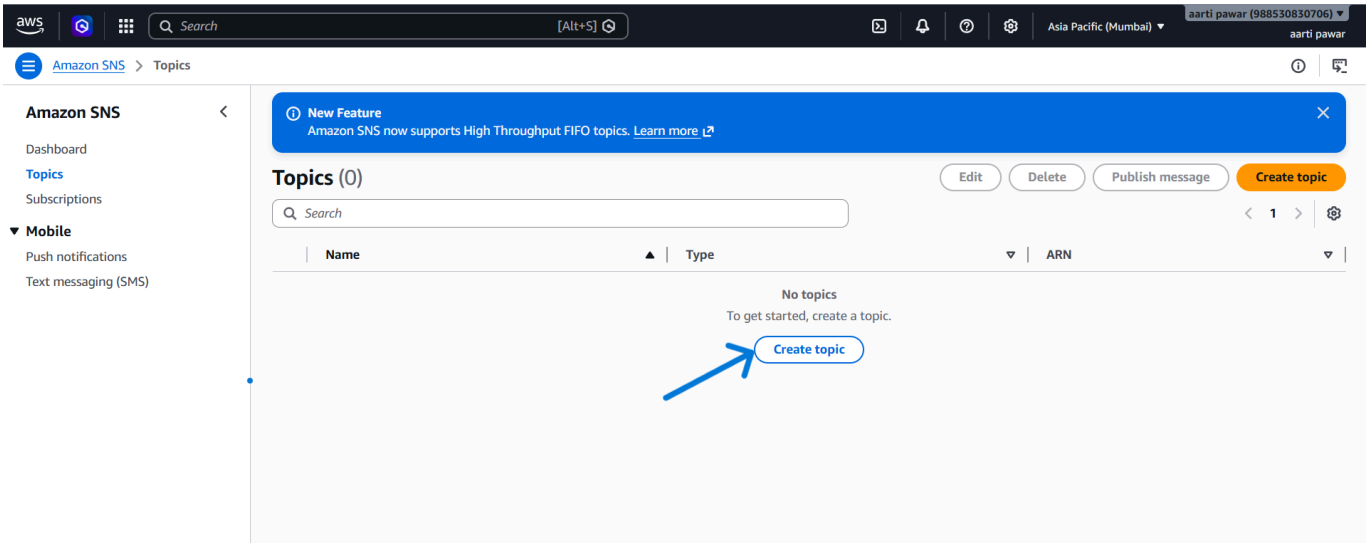


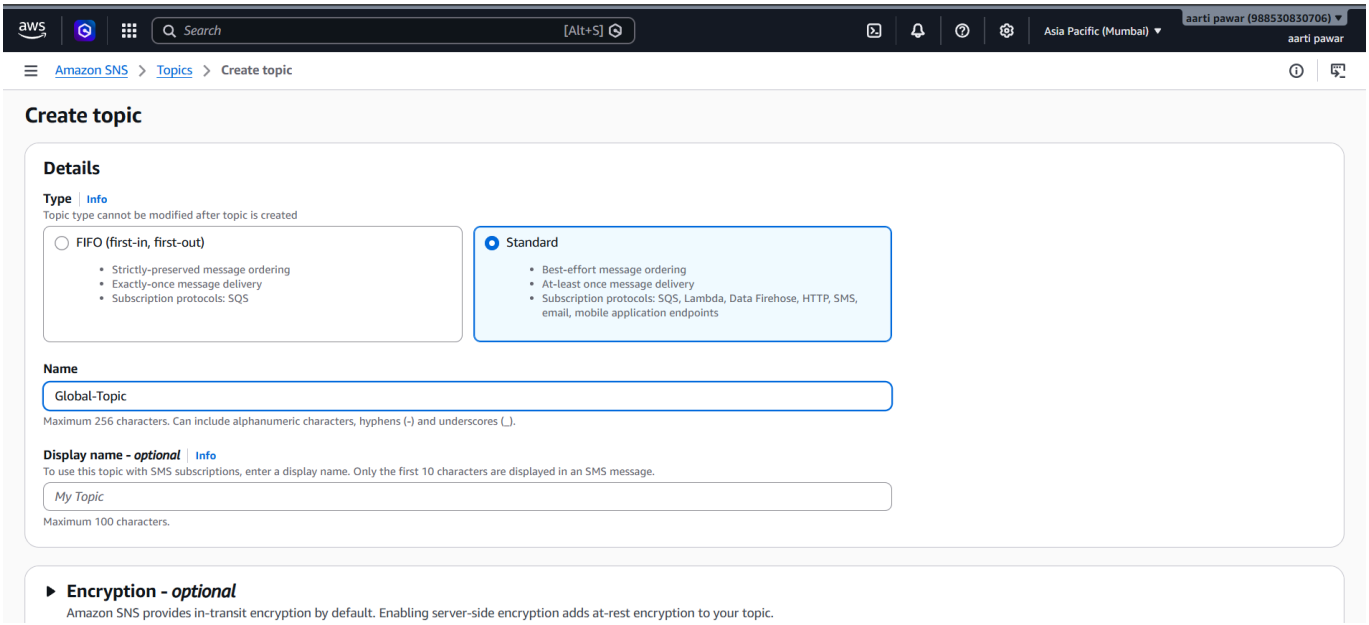
When your alarm is trigger then you get mail from SNS

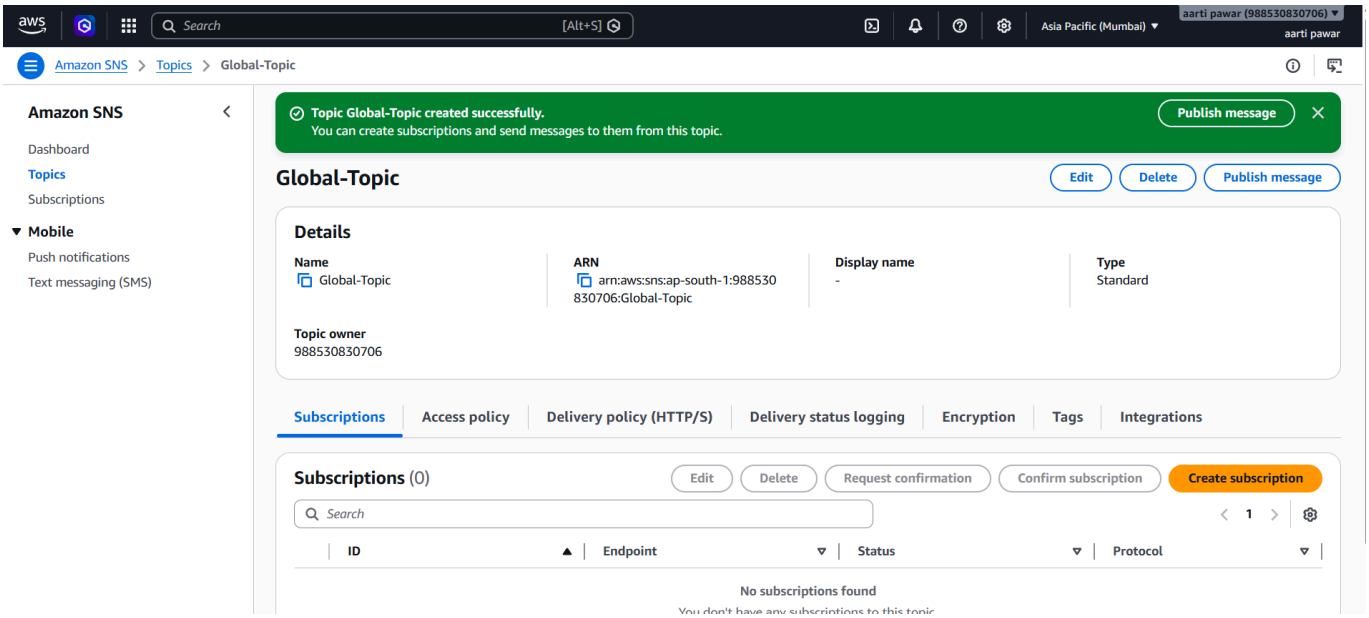




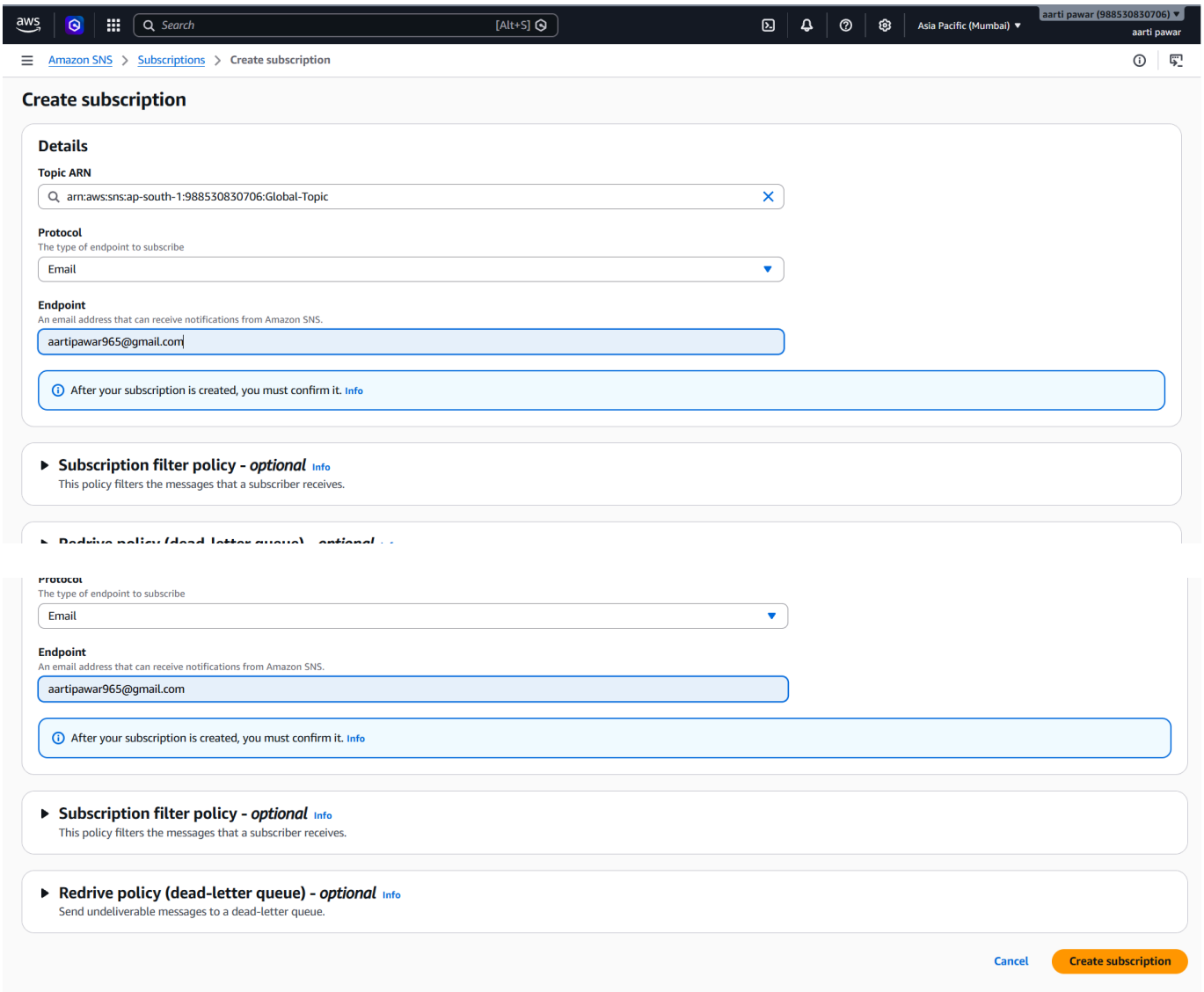
Again create another global topic

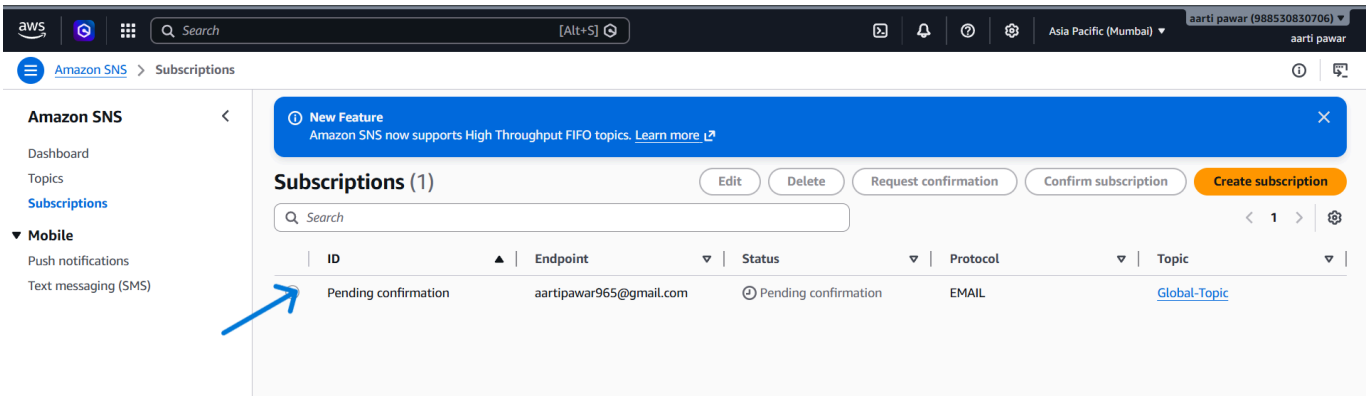


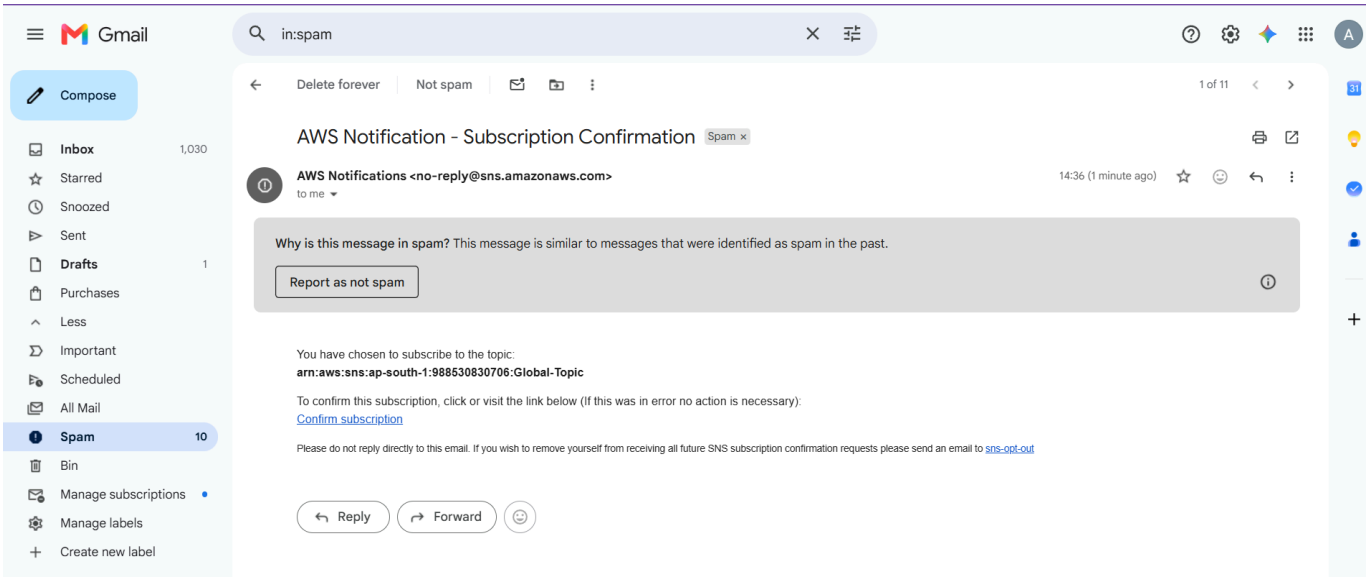


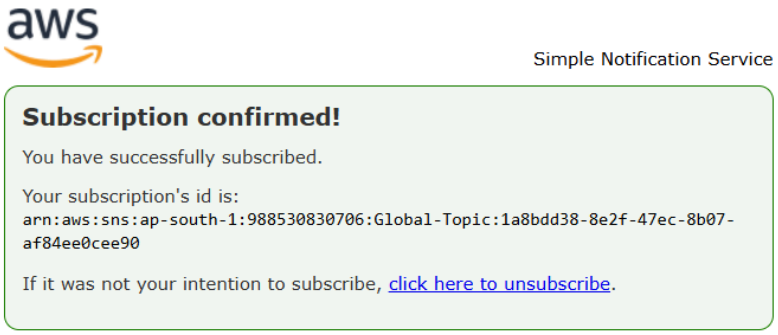


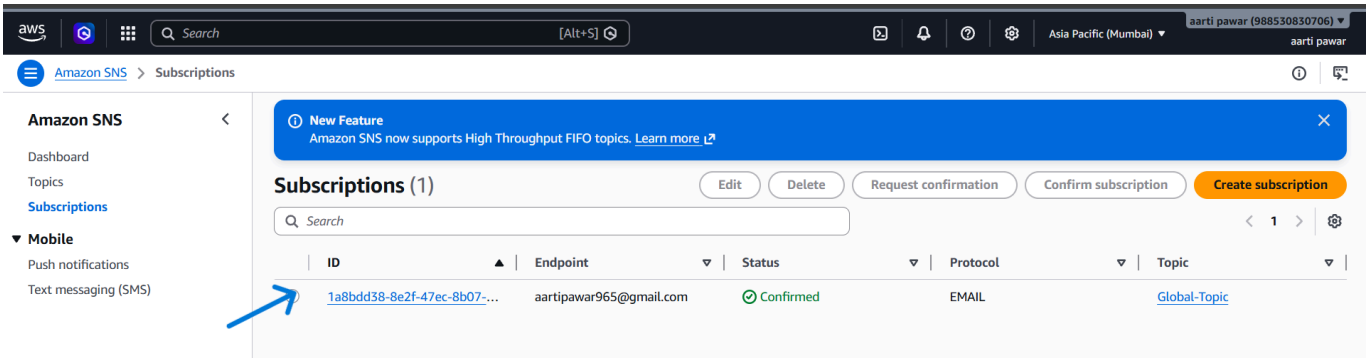
Create Subscription and confirm it from email



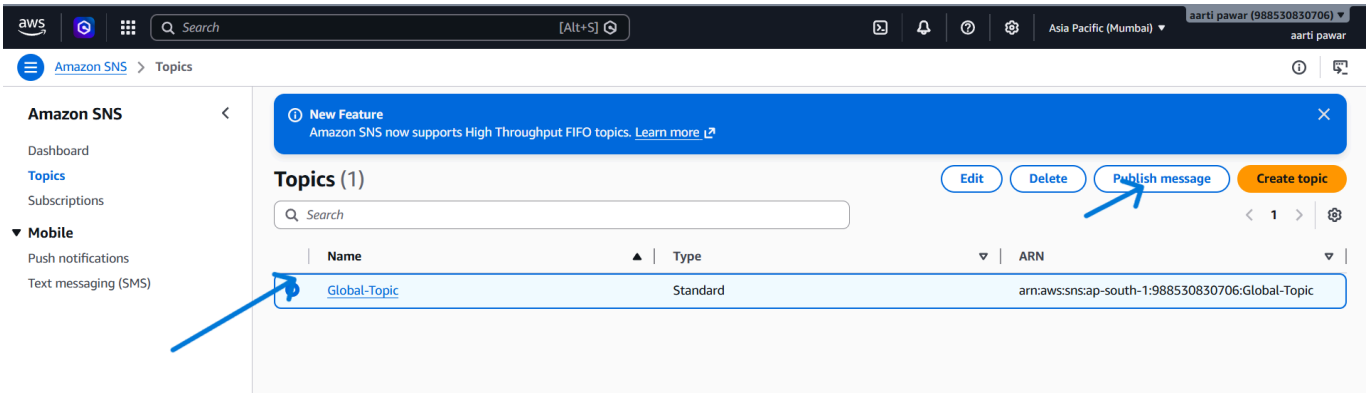


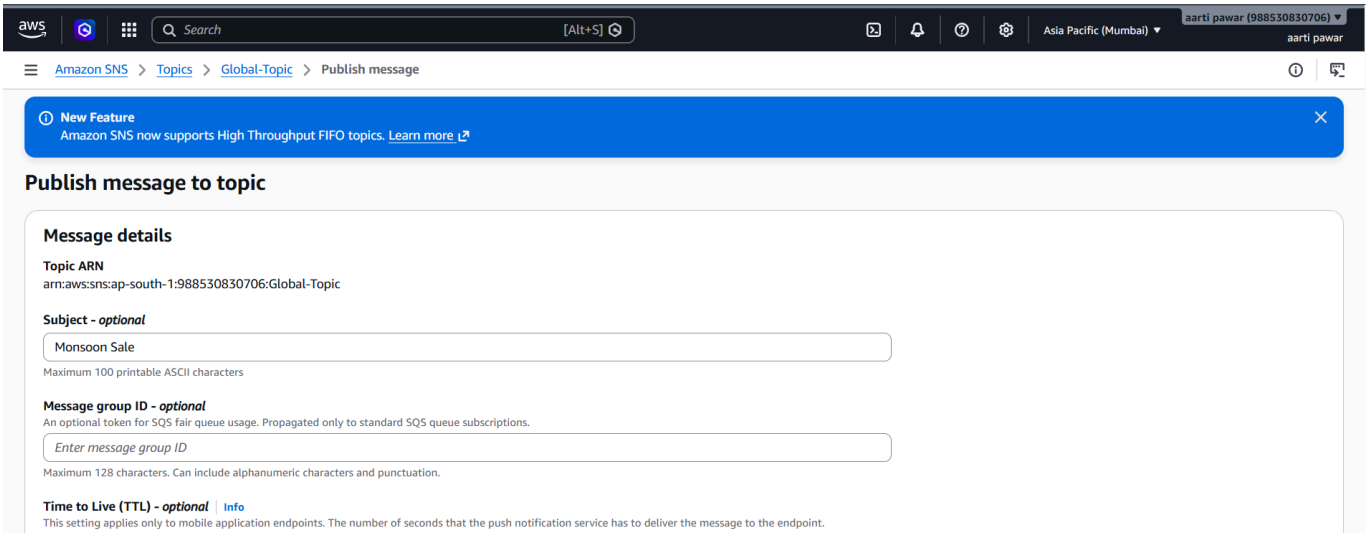


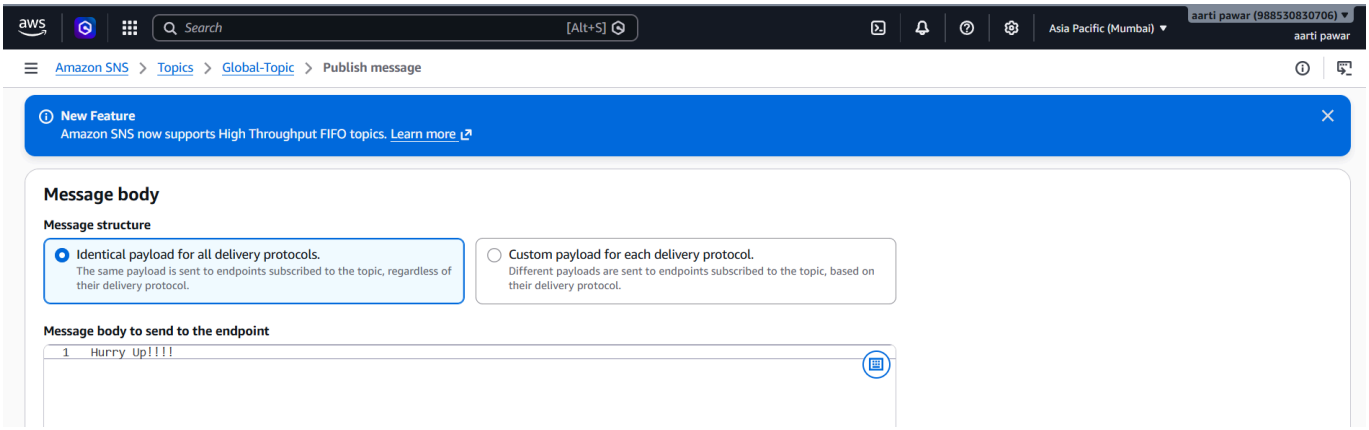


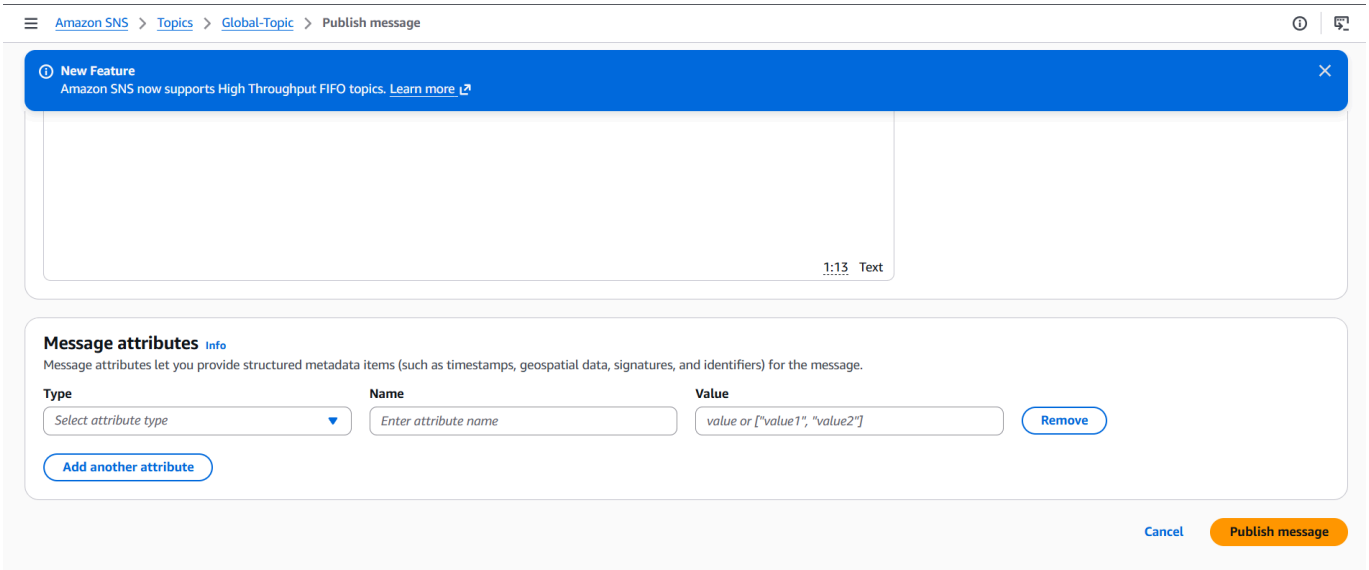


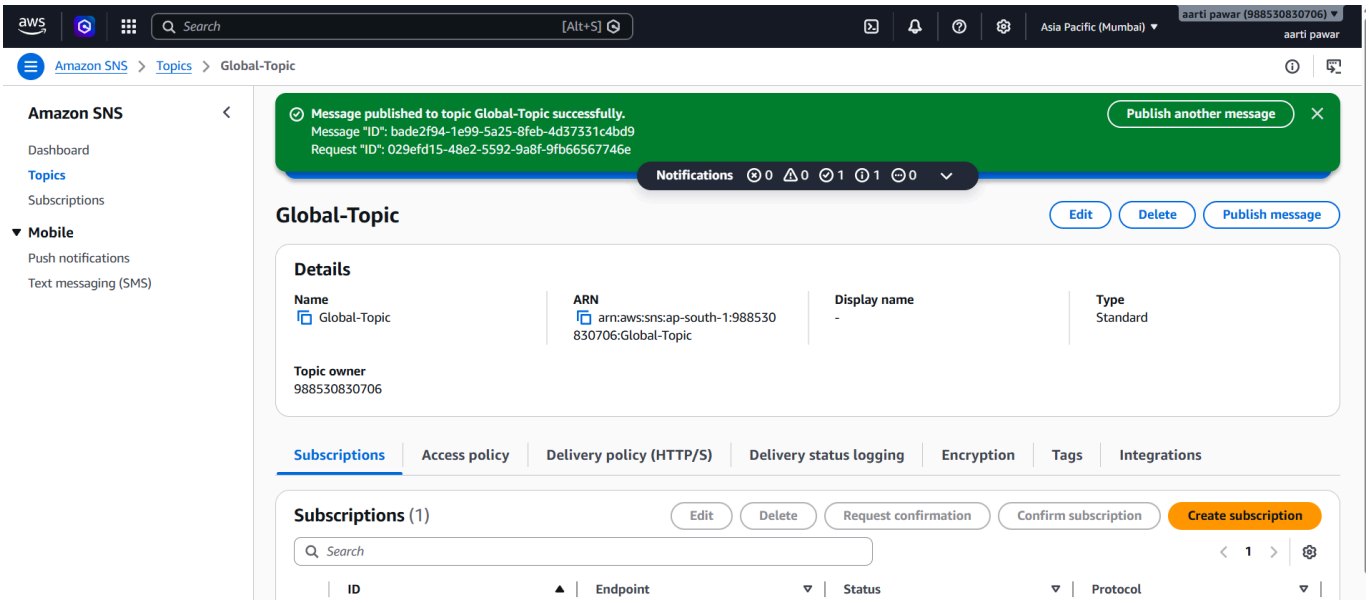
Publish message (Monsoon Sale)


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aarti pawar
Amazon SNS > Topics
Amazon SNS
Dashboard
Topics
Subscriptions
Mobile
Push notifications
Text messaging (SMS)
New Feature
Amazon SNS now supports High Throughput FIFO topics. Learn more
Topics (1)
Edit
Delete
Publish message
Create topic
Search
Name
Type
ARN
Global-Topic
Standard
arn:aws:sns:ap-south-1:988530830706:Global-Topic

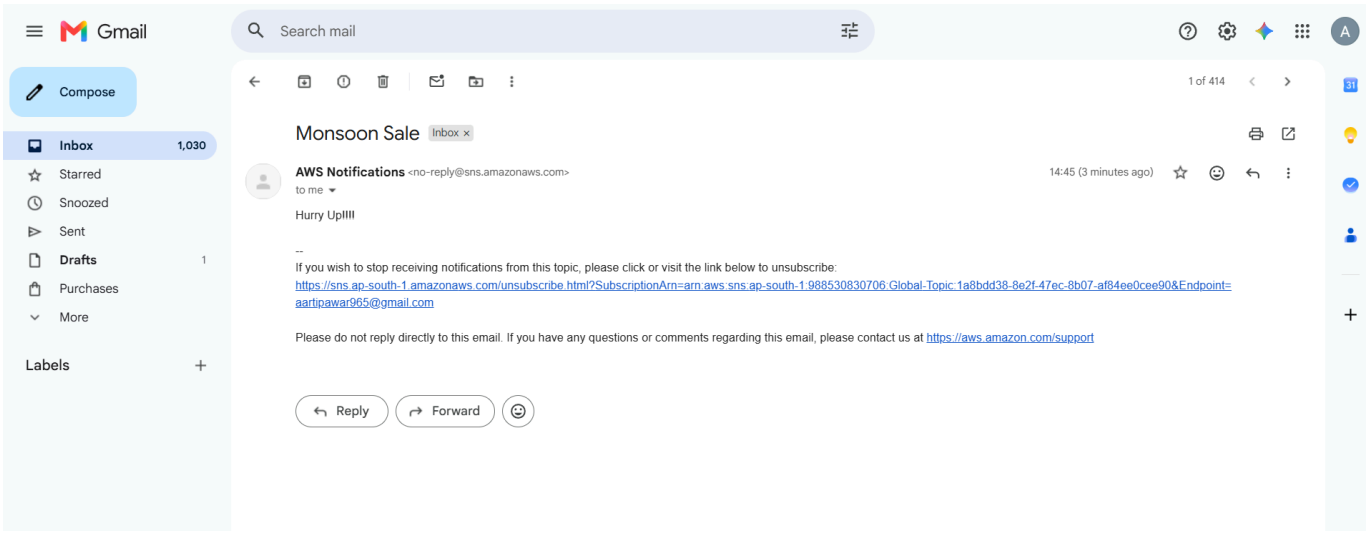

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aarti pawar
Amazon SNS > Topics > Global-Topic > Publish message
New Feature
Amazon SNS now supports High Throughput FIFO topics. Learn more
Publish message to topic
Message details
Topic ARN
arn:aws:sns:ap-south-1:988530830706:Global-Topic
Subject - optional
Monsoon Sale
Maximum 100 printable ASCII characters
Message group ID - optional
An optional token for SQS fair queue usage. Propagated only to standard SQS queue subscriptions.
Enter message group ID
Maximum 128 characters. Can include alphanumeric characters and punctuation.
Time to Live (TTL) - optional Info
This setting applies only to mobile application endpoints. The number of seconds that the push notification service has to deliver the message to the endpoint.


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Amazon SNS > Topics > Global-Topic > Publish message
New Feature
Amazon SNS now supports High Throughput FIFO topics. Learn more
Message body
Message structure
Identical payload for all delivery protocols.
The same payload is sent to endpoints subscribed to the topic, regardless of their delivery protocol.
Custom payload for each delivery protocol.
Different payloads are sent to endpoints subscribed to the topic, based on their delivery protocol.
Message body to send to the endpoint
1 Hurry Up!!!!


Amazon SNS > Topics > Global-Topic > Publish message
New Feature
Amazon SNS now supports High Throughput FIFO topics. Learn more
1:13 Text
Message attributes Info
Message attributes let you provide structured metadata items (such as timestamps, geospatial data, signatures, and identifiers) for the message.
Type
Select attribute type
Name
Enter attribute name
Value
value or ["value1", "value2"]
Remove
Add another attribute
Cancel
Publish message



After that you get monsoon sale notification on email



Features

- Real-time EC2 monitoring
- CloudWatch Agent configuration
- Custom Dashboard visualization
- CPU utilization monitoring
- Memory and Disk monitoring
- Network traffic monitoring
- Automated CloudWatch Alarms
- SNS Email Notifications
- CPU Stress Testing
- End-to-end monitoring validation

Project Workflow

- Launch Amazon EC2
- Configure IAM Role
- Install CloudWatch Agent

- Configure Systems Manager Parameter Store
- Publish EC2 Metrics
- Create CloudWatch Dashboard
- Configure CloudWatch Alarm
- Create SNS Topic
- Confirm Email Subscription
- Execute CPU Stress Test
- Trigger Alarm
- Receive Email Notification

✅ Project Outcome

Successfully implemented a complete AWS monitoring and alerting solution capable of collecting detailed EC2 performance metrics, visualizing infrastructure health through CloudWatch Dashboards, detecting abnormal CPU usage using CloudWatch Alarms, and automatically notifying administrators through Amazon SNS email alerts. This project demonstrates real-world cloud monitoring practices used in production environments.